

QuaRK: A Resource-Aware Quantum Reservoir Kernel for Temporal Learning

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Abstract

Temporal learning requires representations that retain useful history while keeping statistical dependence and implementation cost explicit. We introduce QUARK, a quantum-reservoir-based framework in which a finite-set Johnson–Lindenstrauss projection maps inputs to a quantum register, a bank of reset-contractive reservoirs processes each temporal window, local-observable expectations form a classical feature vector, and a regularized Matérn kernel provides the readout. The analysis follows this pipeline. Exact trace-norm contraction yields a unique causal state and a geometrically decaying finite-window approximation; Matérn RKHS geometry transfers this error to predictions and squared loss. Conditional on distinct exact features, the readout admits a finite-norm interpolant. For a fixed exact featurizer and stationary beta-mixing input–output data, an independent-block argument gives a one-sided risk bound governed by an explicit effective block count and the finite-window error. A finite-family extension makes this guarantee simultaneous over a predeclared collection of featurizer–readout configurations, adding only a logarithmic model-selection term. A separate classical-shadow result quantifies the measurements needed for local Pauli features and propagates finite-shot error uniformly over a bounded RKHS readout class. Together, these results provide a resource-aware, assumption-explicit basis for evaluating quantum reservoir kernels without claiming quantum advantage, hardware-noise robustness, or superiority over classical temporal models.

1 Introduction

Temporal learning is difficult for two related reasons. First, a prediction at one time may depend on events far in the past, so the learner needs a state that summarizes history without growing with the sequence length. Second, training examples extracted from a single trajectory are generally dependent, so their statistical behavior cannot be read directly from an i.i.d. analysis. Reservoir computing addresses the first difficulty by driving a fixed dynamical system with the input sequence and training only a comparatively simple readout (Jaeger, 2001; Maass et al., 2002). When the state dynamics are contractive, the effect of an arbitrary initialization vanishes, yielding a well-defined causal representation of the input history.

Quantum reservoir computing replaces the classical recurrent state by a quantum system and constructs classical features from measurements of the evolved state (Fujii & Nakajima, 2017; Chen et al., 2020). This offers a structured way to generate nonlinear temporal features while keeping the trainable component classical. It also makes the resource picture more layered than in an abstract reservoir model. Input dimension determines how many coordinates must be encoded, reservoir width and circuit depth determine the cost of one state evolution, the observable family determines the feature dimension, and finite-shot estimation determines how often the same temporal window must be replayed. These quantities should be analyzed separately rather than summarized by a single claim of scalability.

A second source of ambiguity is that several distinct questions are often compressed into one performance claim. A contractive state update can guarantee loss of initialization dependence without guaranteeing that different input histories produce different features. A positive-definite kernel can interpolate distinct feature vectors without establishing that the featurizer is injective. Likewise, an efficient estimator for many local

observables does not by itself provide a learning guarantee under temporal dependence or a model of device noise. A useful theory for quantum reservoir learning should therefore connect these layers while preserving their individual assumptions.

We develop this perspective through QUARK, a quantum-reservoir kernel for temporal learning. Each input vector is first mapped to the width of an n -qubit register by a finite-set Johnson–Lindenstrauss (JL) projection and a bounded angle encoding. A collection of independently parameterized reset-contractive quantum channels then processes a temporal window. Local observable expectations from the resulting reservoir states are concatenated into a classical feature vector, on which a regularized Matérn kernel readout is trained. The components have complementary roles: the projection controls encoded width, contraction controls the influence of the remote past, reservoir multiplexing and observable choice control feature dimension, and RKHS regularization controls the readout class.

The analysis follows the same order as the learning pipeline. We first account for the finite-set projection dimension, then use exact trace-norm contraction to construct a unique causal state and quantify the error introduced by a finite window. The geometry of the Matérn RKHS transfers this state-space error to predictions and squared loss. We next characterize the finite-sample capacity of the readout conditional on distinct exact features. Finally, an independent-block argument gives a one-sided risk bound for stationary beta-mixing input–output data. We extend this result from one fixed design to a predeclared finite family, so validation-based selection among finitely many reservoir and readout configurations incurs an explicit logarithmic penalty. A separate classical-shadow result quantifies the measurements needed to estimate local Pauli features. Keeping model selection and measurement separate from the exact-feature risk bound makes the scope of each result explicit.

Throughout the paper, *resource-aware* means that logical width, temporal channel applications, state preparations, observable locality, classical projection work, feature dimension, and kernel-training costs are reported as separate resources. It is not used as a synonym for computational advantage or practical efficiency.

Contributions. The main contributions are:

1. We formulate an end-to-end temporal learner built from reset-contractive quantum reservoirs and a classical Matérn kernel readout. Spatial multiplexing is represented by a tuple of independently evolved reservoir states followed by concatenated observable features, which keeps peak width and repeated execution costs distinct.
2. We prove exact trace-norm contraction of the reset dynamics, derive a unique causal state and a geometrically decaying finite-window feature error, and propagate this approximation through the Matérn RKHS to prediction and squared loss.
3. We separate readout capacity from statistical generalization. Conditional on pairwise-distinct exact features, strict positive definiteness of the Matérn kernel yields a finite-norm interpolant. For a fixed exact featurizer and stationary beta-mixing data, a corrected independent-block analysis yields a one-sided risk bound with an explicit effective block count and finite-window approximation term. A finite-family corollary makes the bound simultaneous over a predeclared collection of configurations and quantifies the additional model-selection cost.
4. We state the local-Pauli classical-shadow requirement independently of the learning bound and derive how coordinate-wise measurement error affects evaluation of a fixed RKHS readout.

The resulting contribution is an assumption-explicit foundation for studying quantum-reservoir-based temporal features. The theory does not establish quantum advantage or superiority over classical reservoirs; those are comparative empirical questions that require matched protocols.

Organization. After positioning the work in Section 2, we formalize temporal windows, reservoir features, risks, and dependence in Section 3. Section 4 then builds the complete learner and accounts for its resources. The theoretical analysis is developed in Section 5, with proofs in Section B. The empirical section then

specifies the controlled functional hierarchy, dependent-family robustness study, high-dimensional width-control experiment, matched controls, and measurement ablations. The exact synthetic generators and task maps are collected in Sections C and D, followed by the discussion and conclusion.

2 Related Work

QUARK sits at the intersection of reservoir computing, quantum feature generation, kernel learning, and statistical learning from dependent data. The method does not replace the central ideas from these areas; rather, it combines them in a temporal architecture whose dynamical, statistical, and measurement assumptions can be tracked jointly.

Reservoir computing, causal filters, and fading memory. Echo-state networks and liquid-state machines established the principle of driving a fixed recurrent system and training only its readout (Jaeger, 2001; Maass et al., 2002). Fading memory formalizes continuity of a causal filter with respect to increasingly remote inputs (Boyd & Chua, 1985), while universality and risk analyses study, respectively, the functions representable by reservoir families and the statistical behavior of learned readouts (Grigoryeva & Ortega, 2018; Gonon et al., 2020). In the present work, uniform contraction is used for the narrower purpose of proving loss of initialization dependence and uniqueness of the causal state for one reset-based architecture. We do not rely on a separate universality or formal fading-memory theorem.

Quantum reservoir computing. Quantum reservoir computing realizes the driven state system with quantum dynamics (Fujii & Nakajima, 2017). Existing proposals have considered dissipative evolution, reset mechanisms, noisy processors, weak or repeated measurements, and spatial multiplexing (Chen & Nurdin, 2019; Chen et al., 2020; Nakajima et al., 2019; Molteni et al., 2023; Mujal et al., 2023; Hu et al., 2024). Finite-dimensional quantum reservoir filters and the role of input dependence have also received formal treatment (Martínez-Peña & Ortega, 2023). QUARK uses a replacement channel whose contraction coefficient is exact and tunable. Its multiple reservoir instances are modeled as separate state evolutions, so they may be executed in parallel or sequentially on reused hardware; this distinction is important for separating peak width from runtime and state-preparation cost.

Quantum features and kernel readouts. Quantum feature methods encode classical data into quantum states or measurement statistics before applying a classical learner (Schuld & Killoran, 2019; Huang et al., 2021). Here the features are history-dependent: they are generated after a quantum reservoir has processed a temporal window. The downstream Matérn kernel is an ordinary classical positive-definite kernel on these measured features. Kernel ridge regression provides a regularized nonlinear readout, while the normalized RKHS structure bounds predictions and allows feature perturbations to be propagated to the readout (Genton, 2001; Porcu et al., 2024; Hofmann et al., 2008). Thus the quantum component lies in the temporal featurizer, not in a claim that the Matérn kernel lacks a classical analogue.

Projection and measurement. The JL lemma preserves pairwise geometry on a finite set under a random linear map whose target dimension grows logarithmically with the set size (Dasgupta & Gupta, 2003; Woodruff, 2014). We use it to relate the number of encoded coordinates to the finite collection of points being processed; the classical projection cost and the subsequent nonlinear dynamics remain part of the method. On the measurement side, classical shadows estimate many observables from shared randomized measurements (Huang et al., 2020). For local random Pauli measurements, a Pauli string of weight at most k has squared shadow norm scaling as 3^k , which determines the locality dependence in our shot bound.

Learning with dependent observations. Independent-block methods compare a beta-mixing sequence with a collection of independent blocks and yield uniform learning bounds in terms of an effective block count (Yu, 1994; Mohri & Rostamizadeh, 2008; Dedecker et al., 2007). We apply this machinery to strided temporal windows. The resulting confidence condition depends on the raw-time separation between blocks, and the stochastic terms scale with the number of approximately independent blocks rather than the raw number of windows.

The standard ingredients above become useful for QUARK only after they are connected to the same temporal feature map. The technical contribution is therefore their architecture-specific integration: tuple-based multiplexing, finite-window error propagation through the Matérn readout, a window-process mixing calculation that retains the effective block count, and separate accounting of exact-feature learning and finite-shot measurement. We now introduce the stochastic objects needed to state this connection precisely.

3 Problem Setting

The analysis distinguishes three objects that coincide only in idealized settings: the causal feature of an infinite input history, the implementable feature obtained from a finite window, and the empirical sample of dependent windows used to train the readout. This section fixes these objects before the quantum architecture is specified.

3.1 Temporal data and windowed observations

Let $Z = (Z_t)_{t \in \mathbb{Z}}$, with $Z_t = (X_t, Y_t)$, be a stationary stochastic process on $\mathbb{R}^d \times \mathbb{R}$. We write $x_{-\infty:0} = (\dots, x_{-1}, x_0)$ for a semi-infinite input history and

$$x_{-w+1:0} = (x_{-w+1}, \dots, x_0) \in (\mathbb{R}^d)^w$$

for its most recent w inputs. A temporal predictor maps a history to a scalar. The target need not have finite memory; a finite window is instead an implementable approximation whose error will later be controlled by reservoir contraction.

For a window length $w \geq 1$ and stride $s = w + g$, with $g \geq 0$, define

$$W_j := (X_{js-w+1:js}, Y_{js}), \quad j \in \mathbb{Z}. \quad (1)$$

The observed sample is $\mathcal{D}_N = (W_1, \dots, W_N)$. The gap g controls the raw-time separation between consecutive windows, but nonoverlap alone does not make them independent.

3.2 Causal and finite-window reservoir features

There are R independently parameterized sub-reservoirs. Sub-reservoir r has an n -qubit state $\rho_t^{(r)}$ and an input-dependent channel $T_{r,x}$. For a semi-infinite history, its causal state at time zero is written $\rho_{\infty,0}^{(r)}(x_{-\infty:0})$. For a finite window, the recursion starts at time $-w$ from a fixed state $\rho_{\text{init}}^{(r)}$, producing $\rho_{w,0}^{(r)}(x_{-w+1:0})$ after the last input is consumed.

Let $\mathcal{O}$ be a finite family of Hermitian observables with $\|O\|_{\infty} \leq 1$. The exact infinite-history and finite-window feature maps are

$$\Phi_{\infty}(x_{-\infty:0}) := (\text{Tr}[O\rho_{\infty,0}^{(r)}])_{r \in [R], O \in \mathcal{O}}, \quad (2)$$

$$\Phi_w(x_{-w+1:0}) := (\text{Tr}[O\rho_{w,0}^{(r)}])_{r \in [R], O \in \mathcal{O}} \in \mathbb{R}^p, \quad p = R|\mathcal{O}|. \quad (3)$$

These vectors concatenate expectations from a tuple of independently evolved states; no tensor-product state is introduced for multiplexing. A finite-shot estimate $\hat{\Phi}_w$ is a distinct random object and will be treated separately from the exact feature map.

3.3 Kernel readout and the three risks

Let $\kappa : \mathbb{R}^p \times \mathbb{R}^p \rightarrow \mathbb{R}$ be a normalized Matérn kernel and $\mathcal{H}_{\kappa}$ its RKHS. For a radius $\Lambda > 0$, define

$$\mathcal{H}_{\kappa,\Lambda} := \{h \in \mathcal{H}_{\kappa} : \|h\|_{\mathcal{H}_{\kappa}} \leq \Lambda\}.$$

Normalization gives $\kappa(z, z) = 1$, and hence

$$|h(z)| \leq \|h\|_{\mathcal{H}_{\kappa}} \sqrt{\kappa(z, z)} \leq \Lambda, \quad h \in \mathcal{H}_{\kappa,\Lambda}. \quad (4)$$

For squared loss $\ell(\hat{y}, y) = (\hat{y} - y)^2$, the causal target is the infinite-history risk

$$\mathcal{R}_\infty(h) := \mathbb{E}[\ell(h(\Phi_\infty(X_{-\infty:0})), Y_0)] . \quad (5)$$

The finite-window approximation is

$$\mathcal{R}_w(h) := \mathbb{E}[\ell(h(\Phi_w(X_{-w+1:0})), Y_0)] , \quad (6)$$

and the quantity optimized from the observed windows is

$$\hat{\mathcal{R}}_{N,w}(h) := \frac{1}{N} \sum_{j=1}^N \ell(h(\Phi_w(X_{js-w+1:js})), Y_{js}) . \quad (7)$$

The learning analysis will therefore have two parts: a statistical comparison between $\mathcal{R}_w$ and $\hat{\mathcal{R}}_{N,w}$, and a deterministic comparison between $\mathcal{R}_\infty$ and $\mathcal{R}_w$.

3.4 Dependence and analytical scope

For sub-sigma-algebras $\mathcal{A}$ and $\mathcal{B}$, let $\beta(\mathcal{A}, \mathcal{B})$ denote their absolute-regularity coefficient. We use

$$\beta_Z(q) := \sup_{t \in \mathbb{Z}} \beta(\sigma(Z_u : u \leq t), \sigma(Z_u : u \geq t + q)) , \quad q \geq 0, \quad (8)$$

and call the process beta mixing when $\beta_Z(q) \rightarrow 0$.

The primary learning result is first stated for a fixed design. The projection, reservoir parameters, initialization states, observable family, window scheme, and Matérn parameters are chosen independently of the sample on which that result is evaluated; when specialized to kernel ridge regression, its regularization parameter is fixed on the same basis. Random design elements, such as the JL matrix or reservoir seed, may be drawn independently and then conditioned upon. We assume that $x \mapsto V_r(x)$, equivalently $x \mapsto T_{r,x}$, is measurable, that the resulting loss classes are measurable (or admit the usual separable versions), that $|Y_t| \leq \Upsilon$ almost surely for some $\Upsilon > 0$, and that the Matérn kernel is normalized. Results using its spectral second moment require $\nu > 1$. The main risk bound concerns exact observable expectations and is uniform over $\mathcal{H}_{\kappa, \Lambda}$. Corollary 10 later extends this uniformity to sample-dependent selection from a finite family of configurations that is fully declared before the sample is examined, including all random seeds and preprocessing maps. Continuous or adaptively generated hyperparameter families remain outside that extension.

With the data, feature maps, risks, and dependence structure in place, we next specify the concrete QUARK pipeline that produces Φ_w .

4 Method

QUARK turns a temporal window into a kernel input through four stages. The raw observations are first projected and encoded as rotations. Each projected sequence then drives several reset-contractive quantum reservoirs. Local observable expectations from their final states are concatenated, and a Matérn kernel ridge-regression readout is trained on the resulting vectors. We describe the stages in this order and conclude with an end-to-end resource ledger.

4.1 Finite-set projection and angle encoding

For resource accounting, let $\mathcal{P} \subset \mathbb{R}^d$ be the set of all individual time-point vectors appearing in the finite collection of windows to be processed, so $|\mathcal{P}| \leq wN$. Independently of these points, and for a chosen register width n , draw

$$\Pi \in \mathbb{R}^{n \times d}, \quad \Pi_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1/n),$$

and set $z = \Pi x$. Each coordinate is converted to a bounded angle, for example $\theta(z_j) = \pi \tanh(z_j)$, and encoded by a one-qubit $R_y(\theta(z_j))$ rotation.

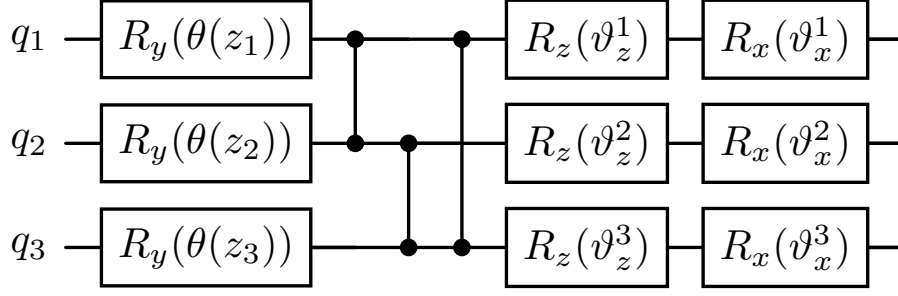


Figure 1: Example input-dependent unitary for a three-qubit sub-reservoir. The controlled links indicate a generic topology-constrained entangling pattern.

The finite-set JL statement in proposition 1 allows the encoded dimension to be chosen from a distortion and failure-probability target rather than by setting $n = d$. This width interpretation applies only to the projected finite set: the dense multiplication still costs $O(nd)$ operations per time point, and the subsequent angle encoding and reservoir evolution are nonlinear. The encoded rotations now provide the input layer of each reservoir update.

4.2 Reset-contractive quantum dynamics

Sub-reservoir r applies the input-dependent unitary

$$V_r(x) = W_r \bigotimes_{j=1}^n R_y(\theta((\Pi x)_j)), \quad (9)$$

where W_r is a fixed topology-constrained unitary. Its parameters are drawn once and then held fixed. A generic three-qubit instance is shown in Figure 1; the controlled links represent a chosen entangling pattern rather than a particular device-native gate set.

After the unitary, the state is partially replaced by a fixed reset state. On an arbitrary operator A , define

$$T_{r,x}(A) := \lambda_r V_r(x) A V_r(x)^\dagger + (1 - \lambda_r) \text{Tr}[A] \rho_{\text{reset}}, \quad 0 \leq \lambda_r < 1. \quad (10)$$

The map $A \mapsto \text{Tr}[A] \rho_{\text{reset}}$ is a replacement channel. A Kraus representation is given in Section A, so it is completely positive and trace preserving; consequently, so is $T_{r,x}$. For a density operator ρ , this reduces to

$$T_{r,x}(\rho) = \lambda_r V_r(x) \rho V_r(x)^\dagger + (1 - \lambda_r) \rho_{\text{reset}}.$$

The parameter λ_r sets the worst-case memory scale: values near one retain more of the previous state, whereas smaller values place more weight on the reset state.

The replacement channel admits the ideal dilation in Figure 2. For $\rho_{\text{reset}} = |+\rangle\langle+|^{\otimes n}$, an n -qubit reset register is prepared in $|+\rangle^{\otimes n}$, and a coin qubit controls swaps between the reservoir and reset registers. Discarding the coin and reset registers produces the mixture in Equation (10).

For a length- w window, the recursion starts from $\rho_{\text{init}}^{(r)}$ and applies Equation (10) once per time point. Moving the initialization into the remote past defines the causal state; lemma 2 will show that this limit exists and is unique.

4.3 Reservoir multiplexing and observable features

The R sub-reservoirs produce a tuple of final states,

$$(\rho_{w,0}^{(1)}, \dots, \rho_{w,0}^{(R)}),$$

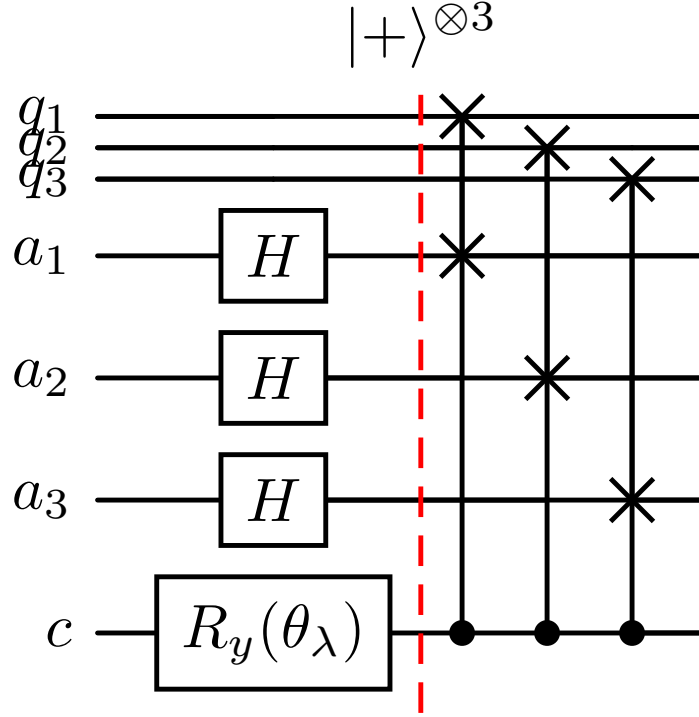


Figure 2: Ideal dilation of the reset channel for three reservoir qubits, with $\theta_\lambda = 2 \arcsin \sqrt{1 - \lambda}$. The reduced channel is obtained by discarding the coin and reset registers.

rather than a tensor-product state. They may be run in parallel on separate registers or sequentially on reused hardware. Parallel execution multiplies peak quantum width, whereas sequential execution keeps the reservoir width at n qubits but multiplies runtime and state preparations.

For each reservoir, the observables in $\mathcal{O}$ are evaluated and concatenated as in Equation (3), producing a vector of dimension $p = R|\mathcal{O}|$. The exact expectations define the analytic feature map used in the kernel and in the main learning theorem. Their finite-shot estimates are introduced later in proposition 12 and corollary 13. Once the quantum states have been reduced to this classical vector, the remaining learning stage is an ordinary kernel method.

4.4 Matérn kernel readout

For distance $r \geq 0$, smoothness $\nu > 1$, and length scale $\xi > 0$, let

$$\varphi_{\nu,\xi}(r) := \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} r}{\xi} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu} r}{\xi} \right), \quad (11)$$

where K_ν is the modified Bessel function of the second kind. With $\varphi_{\nu,\xi}(0) = 1$, define

$$\kappa(z, z') := \varphi_{\nu,\xi}(\|z - z'\|_2). \quad (12)$$

This is a classical kernel evaluated after feature extraction. We use its strict positive definiteness to study interpolation and its spectral second moment to control sensitivity to feature perturbations.

Given exact training features $z_i = \Phi_w(x_i)$, kernel ridge regression solves

$$\min_{h \in \mathcal{H}_\kappa} \frac{1}{N} \sum_{i=1}^N (h(z_i) - y_i)^2 + \eta \|h\|_{\mathcal{H}_\kappa}^2. \quad (13)$$

Table 1: Principal resource parameters of QUARK.

Quantity	Accounting
Encoded reservoir width	n qubits per sub-reservoir
Ideal reset dilation	Up to $2n + 1$ qubits when reset ancillas and coin are simultaneous
Reservoir realizations	R ; sequential reuse or R -fold parallel width
Exact feature dimension	$p = R \mathcal{O} $
Projection work	$O(nd)$ per time point for a dense Π ; values may be cached across snapshots
Per-state temporal evolution	w channel applications, each containing input encoding, W_r , and one reset-channel realization
Finite-shot state preparations	NRS for N windows, R reservoirs, and S snapshots per state
Finite-shot channel updates	$NRSw$; gate count and depth depend on W_r and the reset implementation
Kernel storage/training	$O(N^2)$ memory and $O(N^3)$ naive time

By the representer theorem (Kimeldorf & Wahba, 1971; Schölkopf & Smola, 2002), $\hat{h}_\eta(\cdot) = \sum_i \alpha_i \kappa(z_i, \cdot)$, where

$$\boldsymbol{\alpha} = (K + N\eta I)^{-1} \mathbf{y}.$$

When $|y_i| \leq \Upsilon$, comparing the objective at $\hat{h}_\eta$ with the zero function gives

$$\eta \left\| \hat{h}_\eta \right\|_{\mathcal{H}_\kappa}^2 \leq \frac{1}{N} \sum_{i=1}^N y_i^2 \leq \Upsilon^2, \quad \left\| \hat{h}_\eta \right\|_{\mathcal{H}_\kappa} \leq \frac{\Upsilon}{\sqrt{\eta}}. \quad (14)$$

Thus fixed- η KRR belongs to the RKHS ball used in Section 5 with $\Lambda = \Upsilon/\sqrt{\eta}$, provided the featurizer, kernel parameters, and η are fixed independently of the analyzed sample. The finite-family result in corollary 10 also allows these quantities to be selected from a predeclared finite grid. Dense KRR requires $O(N^2)$ kernel storage and $O(N^3)$ naive linear-algebra time, which remains part of the end-to-end cost.

4.5 Resource summary

The stages above expose different bottlenecks. Table 1 separates peak quantum width from repeated temporal evolution and from classical post-processing. In particular, one snapshot of a final length- w reservoir state requires replaying all w channel updates. Projected inputs may be cached, but the quantum evolution must be repeated for each state preparation.

This decomposition will reappear in the theory: the projection controls encoded width, contraction controls the finite-history approximation, RKHS geometry controls the readout, dependence determines the effective sample size, and observable locality controls the measurement budget.

5 Theory

The method naturally leads to a sequence of analytical questions. How wide must the encoded input be on a finite dataset? Does the reservoir state depend on an arbitrary initialization? How accurately does a finite window represent the causal infinite-history feature? How does this approximation interact with the Matérn readout, how does temporal dependence change the learning rate, and what price is paid when validation selects among finitely many candidate configurations? We answer these questions in the same order as the pipeline and then return to the finite-shot cost of estimating the observable features. Unless stated otherwise, the results use the analytical scope specified in Section 3.4. Complete proofs are collected in Section B.

5.1 Finite-set projection

Proposition 1 (Finite-set JL projection; adapted from Dasgupta & Gupta, 2003; Woodruff, 2014). *Let $\mathcal{P} \subset \mathbb{R}^d$ contain $M \geq 2$ points. For $\varepsilon, \delta \in (0, 1)$, there is a universal constant $C > 0$ such that, if*

$$n \geq C\varepsilon^{-2} \log\left(\frac{M}{\delta}\right)$$

and $\Pi_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1/n)$, then with probability at least $1 - \delta$,

$$(1 - \varepsilon) \|u - v\|_2^2 \leq \|\Pi u - \Pi v\|_2^2 \leq (1 + \varepsilon) \|u - v\|_2^2$$

simultaneously for all $u, v \in \mathcal{P}$. For the points occurring in N windows of length w , one may take $M \leq wN$.

For a prescribed finite set and distortion level, the proposition makes the encoded dimension logarithmic in the number of points rather than tied directly to d . If the point set is the realized sample, the statement is read conditionally on that set, with Π drawn independently. This JL event is separate from the learning-theorem event unless their failure probabilities are combined explicitly. The statement concerns the projection before angle encoding: it neither guarantees that the final reservoir features are distinct nor removes the d -dependent cost of applying Π . Having accounted for encoded width, we next study the temporal state generated from these inputs.

5.2 Contraction, causal states, and finite windows

Lemma 2 (Exact trace-norm contraction). *For every sub-reservoir r , input x , and pair of density operators ρ, ρ' ,*

$$\|T_{r,x}(\rho) - T_{r,x}(\rho')\|_1 = \lambda_r \|\rho - \rho'\|_1. \quad (15)$$

Consequently, for each input history there is a unique causal state obtained as the initialization time tends to $-\infty$. Under the measurability condition in Section 3.4, the causal-state map is measurable. Two trajectories driven by the same last w inputs satisfy

$$\left\| \rho_0^{(r)} - \rho_0'^{(r)} \right\|_1 \leq 2\lambda_r^w. \quad (16)$$

The state-level estimate immediately controls the observable representation used by the learner.

Proposition 3 (Finite-window feature error). *For every input history,*

$$\|\Phi_\infty(x_{-\infty:0}) - \Phi_w(x_{-w+1:0})\|_2 \leq 2\sqrt{|\mathcal{O}| \sum_{r=1}^R \lambda_r^{2w}} \quad (17)$$

$$\leq 2\sqrt{R|\mathcal{O}|} \lambda_{\max}^w, \quad \lambda_{\max} := \max_r \lambda_r. \quad (18)$$

Together, lemma 2 and proposition 3 show that the reservoir has a unique input-determined causal state and that a finite-window feature vector approaches its infinite-history counterpart geometrically. The parameters λ_r therefore expose a worst-case memory scale, while the window length w controls the corresponding approximation error. The next step is to determine how this feature error affects the kernel readout.

5.3 Matérn stability and truncation of loss

Lemma 4 (Matérn RKHS section stability; derived from Genton, 2001; Porcu et al., 2024). *Let κ be the normalized Matérn kernel in Equations (11) and (12) with $\nu > 1$. Then*

$$\|\kappa(z, \cdot) - \kappa(z', \cdot)\|_{\mathcal{H}_\kappa} \leq \frac{1}{\xi} \sqrt{\frac{\nu}{\nu - 1}} \|z - z'\|_2. \quad (19)$$

Corollary 5 (Prediction and loss truncation). *For every $h \in \mathcal{H}_{\kappa, \Lambda}$,*

$$|h(\Phi_\infty) - h(\Phi_w)| \leq \frac{2\Lambda}{\xi} \sqrt{\frac{\nu|\mathcal{O}|}{\nu-1} \sum_{r=1}^R \lambda_r^{2w}}, \quad (20)$$

$$|\mathcal{R}_\infty(h) - \mathcal{R}_w(h)| \leq \tau_w, \quad (21)$$

where

$$\tau_w := \frac{4\Lambda(\Lambda + \Upsilon)}{\xi} \sqrt{\frac{\nu|\mathcal{O}|}{\nu-1} \sum_{r=1}^R \lambda_r^{2w}}. \quad (22)$$

The corollary converts dynamical forgetting into a uniform approximation of predictions and squared risk over the RKHS ball. This term will appear additively in the learning bound. In particular, increasing the sample size alone cannot remove τ_w when w and the reservoir parameters are held fixed.

5.4 Conditional interpolation capacity

The truncation result concerns stability of a bounded readout. A separate question is how much norm is needed to fit a finite set of exact features.

Theorem 6 (Conditional finite-sample interpolation; cf. [Aronszajn, 1950](#); [Kimeldorf & Wahba, 1971](#); [Hofmann et al., 2008](#)). *Let $z_i = \Phi_w(x_i) \in \mathbb{R}^p$, $i \in [N]$, be pairwise distinct exact feature vectors, and let $K_{ij} = \kappa(z_i, z_j)$. The Matérn Gram matrix K is positive definite. If $\mathbf{y} \neq 0$, define*

$$\Lambda_\star := \sqrt{\mathbf{y}^\top K^{-1} \mathbf{y}}.$$

Then there is a unique minimum-norm interpolant $h_\star \in \mathcal{H}_\kappa$ with $\|h_\star\|_{\mathcal{H}_\kappa} = \Lambda_\star$ and $h_\star(z_i) = y_i$ for all i . Therefore:

1. *for $\Lambda \geq \Lambda_\star$,*

$$\min_{\|h\|_{\mathcal{H}_\kappa} \leq \Lambda} \widehat{\mathcal{R}}_{N,w}(h) = 0;$$

2. *for $0 \leq \Lambda \leq \Lambda_\star$,*

$$\min_{\|h\|_{\mathcal{H}_\kappa} \leq \Lambda} \widehat{\mathcal{R}}_{N,w}(h) \leq \left(1 - \frac{\Lambda}{\Lambda_\star}\right)^2 \frac{\|\mathbf{y}\|_2^2}{N}. \quad (23)$$

If $\mathbf{y} = 0$, the zero function interpolates with norm zero.

The theorem isolates the capacity of the Matérn readout: once the exact feature vectors are distinct, a finite RKHS norm suffices to interpolate them, and scaling the interpolant gives a constructive training-error bound below that norm. Pairwise distinctness is an assumption of the result; neither the JL projection nor reservoir contraction guarantees it. This distinction is important because interpolation capacity alone says nothing about test risk. We now turn to that statistical question.

5.5 Dependence of strided windows

The learning sample is the window process (W_j) , not the raw process (Z_t) . The following lemma translates their mixing coefficients by tracking the first raw-time index appearing in a future window.

Lemma 7 (Mixing coefficient of the window process). *Let W be the process in Equation (1) with stride $s = w + g$. For every integer $a \geq 1$,*

$$\beta_W(a) \leq \beta_Z(as - w). \quad (24)$$

In particular, $\beta_W(1) \leq \beta_Z(g)$.

The first raw input in a window at lag a occurs at $as - w + 1$. The displayed bound is conservative by one time index under Equation (8), using monotonicity of beta-mixing coefficients. It supplies the dependence term needed by the independent-block construction.

5.6 Generalization with dependent windows

For a sample $\mathcal{S}_\mu = (\widetilde{W}_1, \dots, \widetilde{W}_\mu)$, define the empirical Rademacher complexity using the convention

$$\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}) := \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \left| \frac{2}{\mu} \sum_{j=1}^{\mu} \sigma_j f(\widetilde{W}_j) \right| \right], \quad \sigma_j \stackrel{\text{i.i.d.}}{\sim} \text{Unif}\{-1, +1\}. \quad (25)$$

This is the absolute-value convention used by [Mohri & Rostamizadeh \(2008\)](#); keeping it explicit avoids hidden factor or centering changes when their theorem is specialized. For a length $N = 2\mu a$ sequence of windows, $\mathcal{S}_\mu$ denotes one fixed position from each of the μ odd blocks of length a . Let

$$\mathcal{F}_\Lambda := \{(x_{-w+1:0}, y) \mapsto (h(\Phi_w(x_{-w+1:0})) - y)^2 : h \in \mathcal{H}_{\kappa, \Lambda}\}.$$

Theorem 8 (One-sided risk bound for beta-mixing windows; adapted from [Mohri & Rostamizadeh, 2008](#)). *Under the analytical scope of Section 3.4, suppose that Z is stationary and beta mixing. Let $N = 2\mu a$, with $\mu, a \in \mathbb{N}$, and set*

$$\delta' := \delta - 4(\mu - 1)\beta_Z(as - w). \quad (26)$$

For every $\delta \in (0, 1)$ satisfying $\delta' > 0$, with probability at least $1 - \delta$, simultaneously for all $h \in \mathcal{H}_{\kappa, \Lambda}$,

$$\mathcal{R}_\infty(h) \leq \widehat{\mathcal{R}}_{N,w}(h) + \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\Lambda) + 3(\Lambda + \Upsilon)^2 \sqrt{\frac{\log(4/\delta')}{2\mu}} + \tau_w. \quad (27)$$

Define the source-convention squared-loss complexity constant

$$C_{\Lambda, \Upsilon} := 8\Lambda(\Lambda + \Upsilon) + 2\Upsilon^2. \quad (28)$$

Corollary 9 (Closed-form RKHS bound; using the contraction principle of [Bartlett & Mendelson, 2002](#)). *Under the conditions of theorem 8, with the same probability, simultaneously for all $h \in \mathcal{H}_{\kappa, \Lambda}$,*

$$\begin{aligned} \mathcal{R}_\infty(h) &\leq \widehat{\mathcal{R}}_{N,w}(h) + \frac{C_{\Lambda, \Upsilon}}{\sqrt{\mu}} + 3(\Lambda + \Upsilon)^2 \sqrt{\frac{\log(4/\delta')}{2\mu}} \\ &\quad + \frac{4\Lambda(\Lambda + \Upsilon)}{\xi} \sqrt{\frac{\nu|\mathcal{O}|}{\nu - 1} \sum_{r=1}^R \lambda_r^{2w}}. \end{aligned} \quad (29)$$

The bound decomposes causal true risk into finite-window training risk, two stochastic terms based on the effective block count $\mu = N/(2a)$, and the deterministic truncation term τ_w . Its uniformity over the fixed RKHS ball includes fixed- η KRR through Equation (14), with $\Lambda = \Upsilon/\sqrt{\eta}$. The statement is one-sided, and the condition $\delta' > 0$ makes the dependence-sample-size trade-off explicit: holding $a = 1$ while increasing N need not preserve the confidence condition.

5.7 Selection from a finite candidate family

The preceding bound fixes the featurizer and kernel before observing the sample. Empirical studies, however, commonly compare a finite grid of reservoir and readout configurations and select one using validation performance. The next result makes the bound simultaneous over such a predeclared family.

Corollary 10 (Finite-family model selection; finite-union argument based on [McDiarmid, 1989](#); [Bartlett & Mendelson, 2002](#)). *Let $\mathfrak{C} = \{1, \dots, L\}$, with $L \geq 1$, be a finite, measurable family fixed before observing the sample, including every candidate's random seed and preprocessing map. All candidates share the same window length w , stride s , and block parameters $N = 2\mu a$. Candidate ℓ specifies an exact causal and finite-window featurizer $\Phi_\infty^{(\ell)}, \Phi_w^{(\ell)}$, a normalized Matérn kernel κ_ℓ with $\nu_\ell > 1$, and an RKHS ball*

$$\mathcal{H}_\ell := \{h \in \mathcal{H}_{\kappa_\ell} : \|h\|_{\mathcal{H}_{\kappa_\ell}} \leq \Lambda_\ell\}.$$

Let $\tau_{w,\ell}$ be its truncation term from corollary 5, define

$$B_{\mathfrak{C}} := \max_{\ell \in [L]} (\Lambda_{\ell} + \Upsilon)^2, \quad A_{\mathfrak{C}} := \max_{\ell \in [L]} [8\Lambda_{\ell}(\Lambda_{\ell} + \Upsilon) + 2\Upsilon^2],$$

and retain $\delta' = \delta - 4(\mu - 1)\beta_Z(as - w) > 0$. Then, with probability at least $1 - \delta$, simultaneously for every $\ell \in [L]$ and $h \in \mathcal{H}_{\ell}$,

$$\begin{aligned} \mathcal{R}_{\infty,\ell}(h) &\leq \widehat{\mathcal{R}}_{N,w,\ell}(h) + \frac{A_{\mathfrak{C}}}{\sqrt{\mu}} + 2B_{\mathfrak{C}}\sqrt{\frac{2\log L}{\mu}} \\ &\quad + 3B_{\mathfrak{C}}\sqrt{\frac{\log(4/\delta')}{2\mu}} + \tau_{w,\ell}. \end{aligned} \quad (30)$$

Here $\mathcal{R}_{\infty,\ell}$ and $\widehat{\mathcal{R}}_{N,w,\ell}$ denote the risks induced by candidate ℓ . Consequently, the same event controls any measurable, sample-dependent selector $(\widehat{\ell}, \widehat{h})$ whose output belongs to $\bigcup_{\ell=1}^L \{\ell\} \times \mathcal{H}_{\ell}$.

The additional term scales as $\sqrt{\log L/\mu}$, so validation-based selection among a declared finite grid costs only logarithmically in the number of candidates. The result can cover finite choices of reservoir parameters, random seeds, observable families, and Matérn readout parameters, provided the candidate list and its preprocessing are fixed in advance and the candidates share the window scheme used by the blocking argument. It does not cover continuous optimization, adaptive expansion of the search space, or sample-estimated preprocessing that is not included in the finite family.

For geometric mixing, the block length can be increased slowly enough to control the coupling penalty without reducing the effective sample size too severely.

Corollary 11 (Geometrically mixing case). *Suppose, in addition, that $\beta_Z(q) \leq \beta_0 e^{-cq}$ for constants $\beta_0, c > 0$, and keep w and s fixed. For a compatible sample size $N = 2\mu a$, choose an integer $a \leq N/2$ satisfying*

$$a \geq \left\lceil \max \left\{ 1, \frac{1}{cs} \log \left(\frac{4\beta_0 e^{cw} N}{\delta} \right) \right\} \right\rceil. \quad (31)$$

Then $\delta' \geq \delta/2$. For fixed $\Lambda, \Upsilon, \delta, \beta_0, c, w, s$, the two stochastic terms in Equation (29) are

$$O\left(\sqrt{\frac{\log(N/\delta)}{N}}\right).$$

If the available sample size is not divisible by $2a$, retain $N_a = 2a \lfloor N/(2a) \rfloor$ observations (provided $N_a \geq 2a$); at most $2a - 1$ terminal observations are discarded. For fixed problem parameters, the displayed choice satisfies $a \leq N/2$ for all sufficiently large N .

The corollary recovers an independent-like rate up to a logarithmic factor, but it does not remove the finite-window term. A consistency result would require a joint schedule for w, s, a , and the readout class.

5.8 Finite-shot local-observable estimation

The preceding learning bound treats Φ_w as an exact expectation vector. On a measurement-based implementation, every coordinate must instead be estimated from repeated preparations of the final reservoir state.

Proposition 12 (Local-Pauli classical-shadow requirement; adapted from Huang et al., 2020). *Suppose $\mathcal{O}$ contains $M = |\mathcal{O}|$ nonidentity Pauli strings of weight at most k . In each snapshot, measure every qubit in an independently and uniformly selected X, Y , or Z basis. There is a universal constant $C_{\text{sh}} > 0$ such that*

$$S \geq C_{\text{sh}} 3^k \varepsilon^{-2} \log \left(\frac{2M}{\delta_{\text{sh}}} \right) \quad (32)$$

snapshots of one reservoir state suffice for simultaneous additive error at most ε for all observables, with probability at least $1 - \delta_{\text{sh}}$.

For N windows and R separately prepared sub-reservoir states, the choice

$$S \geq C_{\text{sh}} 3^k \varepsilon^{-2} \log\left(\frac{2MNR}{\delta_{\text{tot}}}\right) \quad (33)$$

gives the same coordinate-wise error simultaneously over all NR states, by a union bound. This requires NRS state preparations and $NRSw$ temporal channel applications. Identity observables, if retained in a feature ledger, are known exactly.

Corollary 13 (Uniform stability of RKHS readouts to measurement error). *Suppose every coordinate of $\hat{\Phi}_w - \Phi_w$ has magnitude at most ε . On this deterministic event, the following inequalities hold simultaneously for every $h \in \mathcal{H}_{\kappa, \Lambda}$:*

$$\left| h(\hat{\Phi}_w) - h(\Phi_w) \right| \leq \frac{\Lambda}{\xi} \sqrt{\frac{\nu R |\mathcal{O}|}{\nu - 1}} \varepsilon, \quad (34)$$

$$\left| \ell(h(\hat{\Phi}_w), y) - \ell(h(\Phi_w), y) \right| \leq \frac{2\Lambda(\Lambda + \Upsilon)}{\xi} \sqrt{\frac{\nu R |\mathcal{O}|}{\nu - 1}} \varepsilon \quad (35)$$

whenever $|y| \leq \Upsilon$.

For the stated measurement ensemble, observable locality determines the exponential factor 3^k , while the number of observables enters only logarithmically. The perturbation corollary is uniform over the RKHS ball and therefore applies to any already-chosen readout in that ball. It compares the same function on exact and estimated features; it does not compare a readout retrained on noisy features with the exact-feature estimator, nor does it cover physical errors in state preparation and evolution. Those effects are separate from finite-shot sampling.

6 Empirical Evaluation

The empirical study is designed as a behavioral characterization of QUARK, rather than as a search for a single favorable benchmark. Its main component uses controlled synthetic processes for which the temporal mechanism, task functional, dependence strength, ambient dimension, and measurement budget can be varied independently. Two smaller components then probe transfer beyond that controlled setting: deterministic chaotic systems, which are deliberately outside the stochastic assumptions of the theory, and three real-world time-series regression datasets used only as external sanity checks. The theory-aligned generators are defined in Section C, the controlled task functionals in Section D, the chaotic stress tests in Section E, and the real-world sanity checks in Section F.

6.1 Empirical questions

Q1: Which temporal computations can the final reservoir features support? We move from one-step prediction to distributed linear memory, smooth second-order memory, delayed recall, and a sparse cross-lag interaction. The resulting hierarchy separates current or near-future information from long memory and nonlinear temporal composition.

Q2: How do task memory and process persistence interact with the reservoir contraction? The delay, memory-decay, and process-persistence parameters are varied independently. This tests whether the preferred contraction setting changes when the data or target requires longer retained history, and whether excessive memory can become detrimental.

Q3: Are the observed strengths and failures stable across distinct dependence mechanisms? Stable vector VARMA, finite-state hidden Markov observations, and Gaussian GARCH(1,1) provide linear recurrence, latent regime persistence, and conditional heteroskedasticity. Shared tasks permit direct comparison across families, while family-specific targets test filtering and volatility-state recovery.

Q4: What happens when the ambient input dimension grows while the quantum input width remains fixed? A dedicated study increases the ambient dimension under both sparse-signal and distributed-signal constructions. It reports prediction, projection distortion, classical projection cost, and width sensitivity. This tests the width-control interpretation of proposition 1; it does not claim end-to-end runtime independence from the original dimension.

Q5: When does QuaRK improve on, match, or underperform classical representations? The principal controls are an echo-state representation, a matched-dimensional random nonlinear map, and a classical JL representation, each followed by the same Matérn readout and comparable validation budget. Raw-window ridge or Matérn regression is retained as a low-cost reference when computationally tractable.

Q6: Which resource choices produce useful operating points? Encoded width, reservoir count, observable locality, and contraction are varied separately. Predictive error is reported together with feature dimension, logical width, reservoir executions, state preparations, and temporal channel applications, so that accuracy gains can be interpreted against their cost.

Q7: How sensitive is the method to finite measurement budgets? Exact features are compared with local-Pauli classical-shadow estimates under both a frozen-readout protocol and a noisy-feature training protocol. The former connects directly to corollary 13; the latter is a practical stress test outside that fixed-readout result.

Q8: Does the architecture remain useful outside the theorem’s data assumptions, and do the controlled findings survive a limited external check? Lorenz-63 and Mackey-Glass trajectories test forecasting under sensitive, nonlinear dynamics without asserting beta mixing. Three heterogeneous real-world TSER datasets provide a small sanity check rather than a claim of broad real-world validation.

6.2 Common protocol and reporting principles

Three empirical tiers. The primary tier consists of stationary beta-mixing synthetic families and bounded targets covered by the analytical assumptions. The second tier uses chaotic dynamical systems whose empirical behavior is explicitly separated from the theorem. The third tier contains three real-world datasets for which stationarity and mixing are not assumed or estimated. Conclusions are labeled according to the tier from which they arise.

Controlled data and task suites. Unless an experiment states otherwise, the synthetic protocol uses five predeclared data-generating seeds, window length $w = 25$, stride $s = 100$, $N_{\text{tr}} = 5000$ chronological training windows, and $N_{\text{te}} = 1000$ chronological test windows. The exact family parameters and burn-in rules are listed in Section C.5; task directions, normalizations, delays, and family-specific targets are fixed in Section D.6. Every theory-aligned synthetic label lies in $[-1, 1]$.

Splits and leakage control. Hyperparameters are selected only within the training portion using a fixed inner chronological training-validation split. The selected candidate is refitted on the complete training portion before the test set is evaluated. True forecasting labels may use future observations, but the corresponding model input always ends at the stated prediction origin. Latent HMM states and GARCH variances are used only to construct synthetic teacher targets. The chaotic and real-world protocols use the chronological or official splits specified in Sections E.3 and F.2.

Predeclared finite candidate families. For each main comparison, the full candidate set is enumerated before validation scores are inspected. A candidate may specify projection and reservoir seeds, n , R , k , λ_r , the observable family, and Matérn parameters (ξ, ν, η) . Candidates covered by corollary 10 share the same window length and stride and use a finite grid with $\nu > 1$. The final report records the grid, its cardinality L , and the selected candidate. Continuous or adaptively expanded searches are labeled as outside the finite-family corollary.

Table 2: Staged empirical design. The full Cartesian product of generators, tasks, resources, and shot counts is intentionally avoided; each stage isolates a distinct scientific question.

Stage	Main setting	Primary purpose
E1	Reference VARMA and functional hierarchy	Capacity, learning curves, and diagnostic task difficulty
E2	VARMA, HMM, and GARCH across persistence levels	Memory-dependence interaction and cross-family robustness
E3	High-dimensional sparse and distributed signals	JL distortion, fixed-width behavior, and dimensional failure modes
E4	Principal classical controls	Representation-level comparative calibration
E5	One-factor architecture sweeps	Accuracy-resource Pareto structure
E6	Exact and finite-shot features	Measurement sensitivity and noisy-feature training
E7	Lorenz-63 and Mackey-Glass	Out-of-scope chaotic stress testing
E8	Three TSER datasets	Limited real-world transfer sanity check

Classical controls. The three principal controls are: (i) an echo-state network with representation dimension matched to $p = R|\mathcal{O}|$; (ii) a matched-dimensional random nonlinear feature map; and (iii) a classical $d \rightarrow n$ JL map. All three use the same Matérn KRR readout and readout-selection grid as QUARK. Raw-window ridge and raw-window Matérn KRR are reported where feasible, but are not described as dimension- or computation-matched controls.

Readout fitting and metrics. An implementation solving $(K + \lambda_K I)\boldsymbol{\alpha} = \mathbf{y}$ uses $\lambda_K = N_{\text{tr}}\eta$ under Equation (13). Controlled tasks report MSE and

$$\text{NRMSE}_{\text{tr}} := \frac{\sqrt{\text{MSE}}}{\widehat{\sigma}_{Y,\text{tr}}}.$$

Per-seed values, means, standard deviations, and paired differences to each principal control are retained. Aggregate ranks or win-tie-loss summaries are never shown without the underlying per-task uncertainty.

Result provenance. Every table and figure is generated from stored artifacts containing the resolved generator, functional, split, model, candidate grid, predictions, and metrics. Failed or incomplete runs remain visible in the audit ledger, and the supplementary material maps each reported number to its script and artifact path.

6.3 Experiment E1: Functional hierarchy and learning behavior

E1 uses the medium-persistence $d = 3$ VARMA realization. With one fixed exact featurizer, a regularization path records training MSE, validation MSE, test MSE, RKHS norm, and Gram-matrix condition number for the future-projection, exponential-memory, and Volterra tasks. A nested-sample study uses

$$N_{\text{tr}} \in \{100, 500, 1000, 2000, 3000, 5000\}$$

with the candidate grid fixed in advance. Its purpose is to characterize learning behavior, not to fit a prescribed $N^{-1/2}$ law. Delayed recall at $\tau \in \{1, 5, 10, 20\}$ and the sparse cross-lag target $(\tau_1, \tau_2) = (0, 15)$ then isolate long-memory and nonlinear interaction requirements. The full target definitions are in Section D.2.

6.4 Experiment E2: Memory, persistence, and dependent-family robustness

E2 varies the memory requirement of both the data and the target. For VARMA, HMM, and GARCH, the low-, medium-, and high-persistence realizations in Table 3 are paired with delayed recall and with exponential-memory targets using a predeclared decay grid. Around the selected architecture, the contraction parameter is varied over

$$\lambda \in \{0.05, 0.1, 0.3, 0.5, 0.8, 0.95\}.$$

The main diagnostic is whether the contraction value minimizing validation error changes systematically with process persistence or target delay, and whether the relationship is non-monotonic.

A complementary cross-family comparison fixes the selected exact-feature configuration and evaluates the exponential-memory and Volterra targets on all three generators. It additionally uses one-step future projection for VARMA, the filtered-regime score for HMM, and the next conditional-volatility state for GARCH. This stage reports both absolute error and paired differences to the principal classical controls; it is not a repeated architecture search for each family.

6.5 Experiment E3: Ambient dimension and JL width control

E3 uses the stable VARMA construction with $d \in \{3, 10, 30, 100, 300, 500\}$ under the sparse and distributed target directions in Section D.4. The main curve fixes $n = 5$, $R = 3$, $k = 2$, and the selected contraction while d grows. A width-sensitivity study uses $n \in \{3, 5, 8\}$ at $d \in \{100, 500\}$.

For every run, we report prediction error, encoded width, feature dimension, projection time, and the empirical distortion statistic

$$\hat{\Delta}_{0.95} := \text{quantile}_{0.95} \left(\left\{ \left| \frac{\|\Pi x - \Pi x'\|_2^2}{\|x - x'\|_2^2} - 1 \right| : (x, x') \in \mathcal{Q} \right\} \right), \quad (36)$$

where $\mathcal{Q}$ is a predeclared sample of training-point pairs. The comparison includes fixed-width QUARK, classical JL plus Matérn KRR, matched random features, and raw-window readouts when tractable. Sparse targets test resistance to nuisance coordinates; distributed targets test whether useful geometry survives compression when signal spans all coordinates.

6.6 Experiment E4: Matched classical calibration

E4 consolidates the principal comparisons from E1–E3 under common splits, feature-dimension reporting, readout grids, and stochastic seeds. For each scenario it reports QUARK, ESN plus Matérn KRR, matched random features plus Matérn KRR, and classical JL plus Matérn KRR. Raw-window models are included as uncompressed references. Feature-dimension matching, identical-readout matching, candidate-count matching, runtime, and memory are reported as separate quantities rather than compressed into a single “matched budget.” The goal is to identify scenario-dependent strengths and failures, not to establish universal dominance.

6.7 Experiment E5: Resource and memory trade-offs

E5 returns to medium-persistence VARMA and performs one-factor sweeps around the selected configuration:

$$n \in \{3, 5, 6, 8, 10\}, \quad R \in \{1, 3, 5, 8\}, \quad k \in \{1, 2, 3\},$$

together with the contraction grid from E2. The Volterra task and delayed recall at $\tau \in \{10, 20\}$ are used because they require nonlinear processing and retained history. Each panel reports predictive error with peak logical width, feature dimension, reservoir executions, exact-feature runtime, and the implied state-preparation and channel-application counts. A Pareto summary identifies non-dominated accuracy–resource operating points.

6.8 Experiment E6: Finite-shot degradation

E6 fixes the selected reservoir and evaluates snapshot counts

$$S \in \{100, 500, 1000, 5000, 10000\}$$

on the exponential-memory, Volterra, and $\tau = 10$ delayed-recall tasks. Each setting is repeated with independent shadow randomness. Two protocols are reported: a readout trained on exact features and evaluated on estimated features, and a readout trained and evaluated on independently estimated features. The exact feature vector is the reference. In addition to task error, we report coordinate-wise feature RMSE,

Euclidean feature error, prediction error, NRS state preparations, and $NRSw$ temporal channel applications. Only the frozen-readout protocol is interpreted through corollary 13.

6.9 Experiment E7: Out-of-scope chaotic stress tests

E7 evaluates Lorenz–63 and Mackey–Glass trajectories under the generators and sampling rules in Section E. Forecast horizons, sampling intervals, and observation-noise levels are varied while keeping chronological splits and predeclared candidate grids. QUARK is compared with the ESN and matched-random-feature controls using the same readout. Forecast error is reported as a function of horizon, together with the horizon at which each method loses a predeclared fraction of the persistence-baseline skill. These experiments are empirical stress tests only: deterministic chaos is not presented as satisfying the stationary beta-mixing assumptions of theorem 8.

6.10 Experiment E8: Limited real-world sanity checks

E8 uses three heterogeneous datasets from the Monash–UEA–UCR time-series extrinsic-regression archive (Tan et al., 2021): Gas Sensor Array Ethanol, Live Fuel Moisture Content, and Hydraulic Systems. They respectively provide a very long univariate sensor trace, a year-long seven-band remote-sensing series, and a short but very high-dimensional multivariate hydraulic record. The audited archive splits, dimensions, preprocessing, and dataset roles are specified in Section F.

The comparison is deliberately small: QUARK, ESN plus Matérn KRR, matched random nonlinear features plus Matérn KRR, and raw-window Matérn KRR when feasible. No stationarity or mixing claim is made for these datasets, and three datasets are not treated as evidence of broad benchmark superiority. Their purpose is to check whether failure modes and relative trends observed in the controlled study persist in naturally collected data. Full dataset-level results belong in the appendix; the main text contains only a compact summary with uncertainty and no aggregate claim beyond this limited transfer check.

Draft status. The empirical objects and protocols above are fixed before the numerical tables are inserted. The final revision will populate E1–E8 from audited artifacts and will update the abstract, discussion, and conclusion only with conclusions supported by those artifacts.

7 Discussion and Limitations

The analysis was designed to keep the dynamical, statistical, and measurement layers of QUARK separate. This modularity clarifies what can be inferred from the current theory and where empirical evidence or additional analysis is still needed.

7.1 Statistical scope and memory trade-offs

The single-design risk bound is uniform over an RKHS norm ball after the projection, reservoirs, observable family, window scheme, kernel parameters, and regularization level have been fixed. Corollary 10 adds a controlled model-selection layer for a predeclared finite family, including sample-dependent validation selection, at a logarithmic cost in the number of candidates. Continuous optimization, adaptively enlarged search spaces, and sample-estimated preprocessing not included in that family still require separate analysis. The assumptions of stationary beta mixing and bounded labels are also substantive; nonstationarity, heavy-tailed targets, and online adaptation lie outside the present result.

The block length a and the window length w play different roles. Increasing a weakens the dependence penalty through $\beta_Z(as - w)$ but reduces the effective block count $\mu = N/(2a)$. Increasing w , or choosing larger contraction parameters, reduces the worst-case loss of remote history, but may increase circuit cost and retain information that is irrelevant to the task. The theory exposes these trade-offs; it does not select an optimal operating point.

7.2 Measurement, hardware, and end-to-end resources

The main learning theorem uses exact observable expectations. The classical-shadow result addresses only finite sampling under independent local Pauli measurements, and corollary 13 propagates this error through a fixed readout. Training with a noisy Gram matrix, coherent gate errors, reset imperfections, decoherence, calibration drift, and correlated measurements require separate treatment. Likewise, the circuit in Figure 2 is an ideal dilation of the replacement channel, not a real-device validation.

Resource accounting is therefore component-wise. The JL projection can reduce the logical encoded width on a finite point set, but applying a dense projection remains d -dependent. The ambient-dimension experiment in Section 6.5 is designed to test fixed-width behavior, empirical projection distortion, and predictive degradation as d grows; it cannot by itself establish end-to-end runtime independence or practical scalability. Sequential execution of the R reservoirs can avoid an R -fold increase in peak width, at the cost of more state preparations and runtime. With N windows and S snapshots per reservoir output, feature extraction requires NRS state preparations and $NRSw$ temporal channel applications. The shadow bound is logarithmic in the number of requested observables for fixed locality, but exponential in the locality k . Dense kernel ridge regression adds quadratic memory and cubic naive training time.

7.3 Comparative interpretation

The state update, reset mechanism, and observable measurement process are quantum, whereas the projection and Matérn readout are classical. None of the interpolation or generalization results identifies a predictive capability that is unavailable to a classical nonlinear reservoir or random feature map. Consequently, the practical value of the quantum featurizer must be assessed through matched comparisons that control feature dimension, readout, data splits, and model-selection effort. We do not claim quantum advantage, asymptotic speedup, state-of-the-art accuracy, or general superiority over classical temporal models.

The chaotic and real-world experiments have a different evidential role from the theory-aligned synthetic study. Lorenz-63 and Mackey-Glass are explicit out-of-scope stress tests, while the three TSER datasets are limited external sanitary checks for which stationarity and beta mixing are not asserted. They may reveal transfer or failure, but they do not expand the formal scope of the risk theorem.

The method is general-purpose and is not tied to a sensitive deployment in its present form. Its foreseeable risks are the ordinary risks of temporal prediction—distribution shift, unreliable extrapolation, and overconfidence—together with overstated resource or advantage claims in an early-stage quantum setting. Making the analytical assumptions and resource ledger visible is intended to reduce these risks rather than to replace application-specific validation.

8 Conclusion

We introduced QUARK, a temporal learning framework that combines a finite-set input projection, reset-contractive quantum reservoirs, concatenated local-observable features, and a regularized Matérn kernel readout. The architecture separates encoded width, reservoir memory, feature dimension, measurement cost, and classical kernel training, allowing each resource to be interpreted on its own terms.

The theoretical analysis mirrors this construction. Exact trace-norm contraction yields a unique causal state and a geometrically decaying finite-window feature error. Matérn RKHS stability transfers that error to predictions and squared risk, while strict positive definiteness characterizes the norm needed to interpolate distinct finite feature sets. For a fixed exact featurizer and stationary beta-mixing data, the independent-block analysis provides a one-sided risk bound governed by an effective block count and the finite-window approximation. A finite-family extension makes the guarantee simultaneous over a predeclared collection of reservoir and readout configurations, thereby covering validation-based selection with an explicit logarithmic penalty. A separate classical-shadow result connects observable locality and measurement accuracy to the evaluation of a fixed readout.

These results establish a coherent analytical basis for studying quantum reservoir kernels, while leaving comparative predictive value to empirical evaluation. The empirical protocol is organized around a predeclared hierarchy of forecasting, fading-memory, delayed-recall, cross-lag, regime-filtering, and volatility tasks; robustness across three beta-mixing data families; an ambient-dimension study of JL width control; matched classical and component controls; resource ablations; finite-shot behavior; out-of-scope chaotic stress tests; three limited real-world sanity checks; and uncertainty across repeated runs. Numerical conclusions will be added only after the corresponding artifacts have been audited.

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A Technical Background

A.1 Quantum states, replacement channels, and trace norm

An n -qubit density operator is a positive semidefinite matrix of trace one on $(\mathbb{C}^2)^{\otimes n}$. The trace norm is $\|A\|_1 = \text{Tr} \sqrt{A^\dagger A}$, and the operator norm is $\|A\|_\infty$. We repeatedly use

$$|\text{Tr}[AB]| \leq \|A\|_\infty \|B\|_1,$$

unitary invariance of the trace norm, and the fact that two density operators have trace-norm distance at most two.

For completeness, the replacement map $\mathcal{R}_\sigma(A) = \text{Tr}[A]\sigma$ is completely positive and trace preserving; this is a finite-dimensional instance of the operator-sum representation of quantum operations (Kraus, 1971). Indeed, if $\sigma = \sum_a r_a |a\rangle\langle a|$ and $\{|b\rangle\}_b$ is an orthonormal basis, the Kraus operators $K_{ab} = \sqrt{r_a} |a\rangle\langle b|$ satisfy

$$\sum_{a,b} K_{ab} A K_{ab}^\dagger = \text{Tr}[A]\sigma, \quad \sum_{a,b} K_{ab}^\dagger K_{ab} = I.$$

Thus the reset update in Equation (10) is a convex combination of two quantum channels.

A.2 Absolute regularity and the independent-block theorem

For sigma-algebras $\mathcal{A}$ and $\mathcal{B}$, their absolute-regularity coefficient can be written as

$$\beta(\mathcal{A}, \mathcal{B}) = \frac{1}{2} \sup_{\{A_i\}, \{B_j\}} \sum_{i,j} |\mathbb{P}(A_i \cap B_j) - \mathbb{P}(A_i)\mathbb{P}(B_j)|,$$

where the supremum ranges over finite measurable partitions from the two sigma-algebras. The coefficient is monotone under inclusion of either sigma-algebra, and measurable transformations cannot increase it (Bradley, 2005).

We use the following direct specialization of Mohri & Rostamizadeh (2008, Theorem 2). Let $(Q_j)_{j \in \mathbb{Z}}$ be stationary and beta mixing, let $\mathcal{F}$ be a fixed measurable class of functions into $[0, B]$, and let $m = 2\mu a$. Divide $Q_1, \dots, Q_m$ into 2μ consecutive blocks of length a , and form $\mathcal{S}_\mu$ by selecting the same fixed coordinate from every odd block. With the source paper’s empirical Rademacher convention

$$\hat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}) := \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \left| \frac{2}{\mu} \sum_{j=1}^{\mu} \sigma_j f(\tilde{Q}_j) \right| \right], \quad (37)$$

if

$$\delta'_Q := \delta - 4(\mu - 1)\beta_Q(a) > 0,$$

then, with probability at least $1 - \delta$, simultaneously for every $f \in \mathcal{F}$,

$$\mathbb{E}[f(Q_0)] \leq \frac{1}{m} \sum_{j=1}^m f(Q_j) + \hat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}) + 3B \sqrt{\frac{\log(4/\delta'_Q)}{2\mu}}. \quad (38)$$

The theorem is one-sided. A two-sided statement would require an additional reverse-deviation argument and a corresponding confidence allocation. We use Equation (37) throughout so that the imported theorem and all constants in our specialization share exactly the same convention.

A.3 Spectral representation of the Matérn kernel

By Bochner’s theorem (Bochner, 1933), the normalized stationary Matérn kernel has a probability spectral measure:

$$\kappa(z, z') = \mathbb{E}_\Omega \left[e^{i\Omega^\top(z - z')} \right] = \mathbb{E}_\Omega \left[\cos(\Omega^\top(z - z')) \right]. \quad (39)$$

For the parameterization in Equation (11), its density on $\mathbb{R}^p$ is

$$p_{\nu, \xi}(\omega) = \frac{\Gamma(\nu + p/2) \xi^p}{\Gamma(\nu)(2\nu\pi)^{p/2}} \left(1 + \frac{\xi^2 \|\omega\|_2^2}{2\nu} \right)^{-(\nu + p/2)}, \quad (40)$$

which is a centered multivariate Student density with 2ν degrees of freedom and scale matrix $\xi^{-2}I$; see Genton (2001); Porcu et al. (2024). It is strictly positive everywhere. For $\nu > 1$, its covariance exists and is

$$\mathbb{E}[\Omega\Omega^\top] = \frac{\nu}{(\nu - 1)\xi^2} I. \quad (41)$$

These identities yield the section-stability constant used in lemma 4. Positivity of the density also yields strict positive definiteness on distinct Euclidean points, as made explicit in the proof of theorem 6; the general RKHS interpolation facts used there are standard (Aronszajn, 1950; Kimeldorf & Wahba, 1971).

A.4 Local Pauli classical shadows

In one snapshot of the local-Pauli shadow protocol, each qubit is measured in an independently and uniformly chosen X , Y , or Z basis. Inversion of the one-qubit measurement channel multiplies a nonidentity Pauli factor by three. Hence a Pauli string of weight q has squared shadow norm 3^q . The median-of-means guarantee of Huang et al. (2020) therefore gives simultaneous estimation of M nonidentity, weight-at-most- k Pauli expectations with $O(3^k \varepsilon^{-2} \log(M/\delta))$ independent snapshots. Identity observables, if included in a feature ledger, are known exactly and require no measurement.

A.5 Provenance of imported ingredients

The replacement-channel representation follows the standard operator-sum framework (Kraus, 1971), and the spectral representation of the stationary Matérn kernel invokes Bochner’s theorem (Bochner, 1933). The finite-set projection result is the standard JL argument (Dasgupta & Gupta, 2003; Woodruff, 2014); the dependent-data inequality is imported from Mohri & Rostamizadeh (2008); the RKHS and representer facts are standard consequences of Aronszajn (1950); Kimeldorf & Wahba (1971); and the finite-union step uses the bounded-difference method of McDiarmid (1989) and standard Rademacher-complexity tools (Bartlett & Mendelson, 2002). The results specific to QUARK are the way these ingredients are composed with its reset-contractive dynamics, finite-window approximation, observable map, and fixed-design temporal learning problem.

B Proofs

B.1 Finite-set JL projection

Proof. For a fixed nonzero vector v , concentration of $\|\Pi v\|_2^2 / \|v\|_2^2$ around one gives

$$\mathbb{P}\left(\left|\|\Pi v\|_2^2 - \|v\|_2^2\right| > \varepsilon \|v\|_2^2\right) \leq 2e^{-c_0 \varepsilon^2 n}$$

for a universal $c_0 > 0$. Apply this inequality to the fewer than $M^2/2$ nonzero differences $u - v$ and take a union bound. Choosing $n \geq C\varepsilon^{-2} \log(M/\delta)$ for a sufficiently large universal C makes the total failure probability at most δ . This is the standard finite-set JL argument (Dasgupta & Gupta, 2003; Woodruff, 2014). If $\mathcal{P}$ is random but independent of Π , the calculation holds conditionally on every realized $\mathcal{P}$, and hence also unconditionally. $\square$

B.2 Trace-norm contraction

Proof. For density operators $\text{Tr}[\rho] = \text{Tr}[\rho'] = 1$, so the replacement components cancel when two outputs with the same input are subtracted:

$$T_{r,x}(\rho) - T_{r,x}(\rho') = \lambda_r V_r(x)(\rho - \rho')V_r(x)^\dagger.$$

Homogeneity and unitary invariance of the trace norm give Equation (15).

Iterating the equality for w common inputs yields

$$\left\|\rho_0^{(r)} - \rho_0'^{(r)}\right\|_1 \leq \lambda_r^w \left\|\rho_{-w}^{(r)} - \rho_{-w}'^{(r)}\right\|_1 \leq 2\lambda_r^w.$$

To construct the causal state, initialize the recursion at time $-m$ in an arbitrary density operator and call the resulting time-zero state $\rho_0^{[m]}$. If $m' > m$, regard the recursion started at $-m'$ as having some density operator at time $-m$. The last m updates are common, so

$$\left\|\rho_0^{[m']} - \rho_0^{[m]}\right\|_1 \leq 2\lambda_r^m.$$

Thus $(\rho_0^{[m]})_m$ is Cauchy in the finite-dimensional trace-norm space and converges. The bound also shows that the limit is independent of the chosen initial states. Shifting the argument in time produces a unique causal state sequence. For every finite start time, the time-zero state is a measurable function of a finite input block. Under Section 3.4, the causal state is their pointwise trace-norm limit and is therefore measurable. $\square$

B.3 Finite-window feature error

Proof. For every r and $O \in \mathcal{O}$, lemma 2 and trace duality give

$$\left|\text{Tr}\left[O(\rho_{\infty,0}^{(r)} - \rho_{w,0}^{(r)})\right]\right| \leq \|O\|_\infty \left\|\rho_{\infty,0}^{(r)} - \rho_{w,0}^{(r)}\right\|_1 \leq 2\lambda_r^w.$$

Squaring and summing over r and O yields

$$\|\Phi_\infty - \Phi_w\|_2^2 \leq 4|\mathcal{O}| \sum_{r=1}^R \lambda_r^{2w}.$$

Taking square roots proves the first inequality. The second follows from $\lambda_r \leq \lambda_{\max}$. $\square$

B.4 Matérn RKHS section stability

Proof. The reproducing property and normalization give

$$\begin{aligned} \|\kappa(z, \cdot) - \kappa(z', \cdot)\|_{\mathcal{H}_\kappa}^2 &= \kappa(z, z) + \kappa(z', z') - 2\kappa(z, z') \\ &= 2(1 - \varphi_{\nu, \xi}(\|z - z'\|_2)). \end{aligned}$$

Let $u = z - z'$. Using the spectral representation in Equation (39), the elementary inequality $1 - \cos t \leq t^2/2$, and Equation (41),

$$\begin{aligned} 2(1 - \kappa(z, z')) &= 2\mathbb{E}[1 - \cos(\Omega^\top u)] \\ &\leq \mathbb{E}[(\Omega^\top u)^2] \\ &= u^\top \mathbb{E}[\Omega \Omega^\top] u \\ &= \frac{\nu}{(\nu - 1)\xi^2} \|u\|_2^2. \end{aligned}$$

Taking square roots proves the claim. $\square$

B.5 Prediction and loss truncation

Proof. For any $h \in \mathcal{H}_{\kappa, \Lambda}$, the reproducing property, lemma 4, and proposition 3 imply

$$\begin{aligned} |h(\Phi_\infty) - h(\Phi_w)| &= |\langle h, \kappa(\Phi_\infty, \cdot) - \kappa(\Phi_w, \cdot) \rangle_{\mathcal{H}_\kappa}| \\ &\leq \Lambda \|\kappa(\Phi_\infty, \cdot) - \kappa(\Phi_w, \cdot)\|_{\mathcal{H}_\kappa} \\ &\leq \frac{2\Lambda}{\xi} \sqrt{\frac{\nu|\mathcal{O}|}{\nu - 1} \sum_{r=1}^R \lambda_r^{2w}}. \end{aligned}$$

This is Equation (20).

For $u, v \in [-\Lambda, \Lambda]$ and $|y| \leq \Upsilon$,

$$|(u - y)^2 - (v - y)^2| = |u - v| |u + v - 2y| \leq 2(\Lambda + \Upsilon) |u - v|.$$

Both predictions are in $[-\Lambda, \Lambda]$ by Equation (4). Apply the last display pointwise with $u = h(\Phi_\infty)$, $v = h(\Phi_w)$, use the prediction bound above, and take expectations. This gives $|\mathcal{R}_\infty(h) - \mathcal{R}_w(h)| \leq \tau_w$. $\square$

B.6 Conditional interpolation

Proof. The Matérn spectral density in Equation (40) is strictly positive on $\mathbb{R}^p$. For any $\mathbf{c} \in \mathbb{R}^N$, Bochner's representation gives

$$\mathbf{c}^\top K \mathbf{c} = \int_{\mathbb{R}^p} \left| \sum_{i=1}^N c_i e^{i\omega^\top z_i} \right|^2 p_{\nu, \xi}(\omega) d\omega.$$

Suppose that the z_i are distinct and $\mathbf{c} \neq 0$. Choose a direction $v \in \mathbb{R}^p$ outside the finite union of hyperplanes $\{v : v^\top(z_i - z_j) = 0\}$, so the scalars $v^\top z_i$ are pairwise distinct. The restriction $t \mapsto \sum_i c_i e^{itv^\top z_i}$ is a one-dimensional exponential polynomial with distinct frequencies and cannot vanish identically unless all

$c_i = 0$. Thus the multivariate exponential polynomial is nonzero at some point and, by continuity, on an open set of positive Lebesgue measure. Since $p_{\nu,\xi} > 0$ everywhere, the integral is strictly positive. Hence K is positive definite and invertible.

Let $\alpha = K^{-1}\mathbf{y}$ and

$$h_\star(\cdot) = \sum_{i=1}^N \alpha_i \kappa(z_i, \cdot).$$

Then $(h_\star(z_1), \dots, h_\star(z_N))^\top = K\alpha = \mathbf{y}$ and

$$\|h_\star\|_{\mathcal{H}_\kappa}^2 = \alpha^\top K\alpha = \mathbf{y}^\top K^{-1}\mathbf{y} = \Lambda_\star^2.$$

Moreover, if h is any other interpolant, then $g := h - h_\star$ satisfies $g(z_i) = 0$ for all i , so g is orthogonal to every kernel section $\kappa(z_i, \cdot)$ and hence to $h_\star$. Therefore $\|h\|_{\mathcal{H}_\kappa}^2 = \|h_\star\|_{\mathcal{H}_\kappa}^2 + \|g\|_{\mathcal{H}_\kappa}^2$, proving that $h_\star$ is the unique minimum-norm interpolant (Aronszajn, 1950; Kimeldorf & Wahba, 1971).

Thus $h_\star$ is feasible for every $\Lambda \geq \Lambda_\star$ and has zero empirical risk. For $0 \leq \Lambda \leq \Lambda_\star$, the function $h_\Lambda = (\Lambda/\Lambda_\star)h_\star$ is feasible and satisfies

$$h_\Lambda(z_i) - y_i = -\left(1 - \frac{\Lambda}{\Lambda_\star}\right)y_i.$$

Its empirical risk is the right-hand side of Equation (23); the minimum over the ball cannot be larger. The case $\mathbf{y} = 0$ is immediate. $\square$

B.7 Mixing of the window process

Proof. By stationarity it is enough to compare the sigma-algebra generated by $(W_j)_{j \leq 0}$ with that generated by $(W_j)_{j \geq a}$. Every raw variable occurring in a past window has time index at most zero, so

$$\sigma(W_j : j \leq 0) \subseteq \sigma(Z_t : t \leq 0).$$

The first raw input appearing in W_a has index $as - w + 1$, and all raw variables in later windows occur no earlier. Therefore

$$\sigma(W_j : j \geq a) \subseteq \sigma(Z_t : t \geq as - w + 1).$$

Monotonicity of $\beta(\cdot, \cdot)$ under sigma-algebra inclusion gives

$$\beta_W(a) \leq \beta_Z(as - w + 1) \leq \beta_Z(as - w),$$

where the last inequality uses that mixing coefficients are nonincreasing. For $a = 1$ and $s = w + g$, $as - w = g$. $\square$

B.8 One-sided dependent-data risk bound

Proof. For $h \in \mathcal{H}_{\kappa, \Lambda}$, define

$$f_h(W_j) = (h(\Phi_w(X_{js-w+1:js})) - Y_{js})^2.$$

By Equation (4) and $|Y_t| \leq \Upsilon$,

$$0 \leq f_h(W_j) \leq B, \quad B = (\Lambda + \Upsilon)^2.$$

The class $\mathcal{F}_\Lambda = \{f_h : h \in \mathcal{H}_{\kappa, \Lambda}\}$ is fixed independently of the sample under the analytical scope of Section 3.4. Apply Equation (38) to the stationary window process. By lemma 7,

$$\beta_W(a) \leq \beta_Z(as - w),$$

Let $\delta'_W = \delta - 4(\mu - 1)\beta_W(a)$ denote the source-theorem remainder. The mixing bound gives $0 < \delta' \leq \delta'_W$, so the source theorem applies; using the smaller δ' only enlarges the confidence term. Consequently, simultaneously for all $h \in \mathcal{H}_{\kappa, \Lambda}$,

$$\mathcal{R}_w(h) \leq \widehat{\mathcal{R}}_{N,w}(h) + \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\Lambda) + 3(\Lambda + \Upsilon)^2 \sqrt{\frac{\log(4/\delta')}{2\mu}}.$$

Finally, corollary 5 gives $\mathcal{R}_\infty(h) \leq \mathcal{R}_w(h) + \tau_w$. Combining the two inequalities proves Equation (27). $\square$

B.9 Closed-form RKHS complexity bound

Proof. Condition on $\mathcal{S}_\mu = ((u_j, y_j))_{j=1}^\mu$, where $u_j = \Phi_w(x_j)$, and let $\mathcal{G}_\Lambda$ be the prediction class induced by $\mathcal{H}_{\kappa, \Lambda}$ on these features. Because the RKHS ball is symmetric,

$$\begin{aligned}\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{G}_\Lambda) &= \mathbb{E}_\sigma \left[\sup_{\|h\|_{\mathcal{H}_\kappa} \leq \Lambda} \left| \frac{2}{\mu} \left\langle h, \sum_{j=1}^\mu \sigma_j \kappa(u_j, \cdot) \right\rangle_{\mathcal{H}_\kappa} \right| \right] \\ &= \frac{2\Lambda}{\mu} \mathbb{E}_\sigma \left\| \sum_{j=1}^\mu \sigma_j \kappa(u_j, \cdot) \right\|_{\mathcal{H}_\kappa} \\ &\leq \frac{2\Lambda}{\mu} \sqrt{\sum_{j=1}^\mu \kappa(u_j, u_j)} = \frac{2\Lambda}{\sqrt{\mu}}.\end{aligned}$$

For each j , define $\psi_j(t) = (t - y_j)^2 - y_j^2$. On $[-\Lambda, \Lambda]$, $\psi_j(0) = 0$ and ψ_j is $2(\Lambda + \Upsilon)$ -Lipschitz. The absolute-value contraction principle (Bartlett & Mendelson, 2002) gives the safe bound

$$\mathbb{E}_\sigma \sup_{h \in \mathcal{H}_{\kappa, \Lambda}} \left| \frac{2}{\mu} \sum_{j=1}^\mu \sigma_j \psi_j(h(u_j)) \right| \leq 4(\Lambda + \Upsilon) \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{G}_\Lambda) \leq \frac{8\Lambda(\Lambda + \Upsilon)}{\sqrt{\mu}}.$$

The source convention contains an absolute value, so the label-only offset must also be retained. By Jensen's inequality and independence of the Rademacher signs,

$$\mathbb{E}_\sigma \left| \frac{2}{\mu} \sum_{j=1}^\mu \sigma_j y_j^2 \right| \leq \frac{2}{\mu} \sqrt{\sum_{j=1}^\mu y_j^4} \leq \frac{2\Upsilon^2}{\sqrt{\mu}}.$$

The triangle inequality therefore yields

$$\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\Lambda) \leq \frac{8\Lambda(\Lambda + \Upsilon) + 2\Upsilon^2}{\sqrt{\mu}} = \frac{C_{\Lambda, \Upsilon}}{\sqrt{\mu}}.$$

Substitution into Equation (27), followed by the definition of τ_w , proves Equation (29). $\square$

B.10 Finite-family model selection

Proof. For candidate ℓ , let

$$\mathcal{F}_\ell := \left\{ (x_{-w+1:0}, y) \mapsto (h(\Phi_w^{(\ell)}(x_{-w+1:0})) - y)^2 : h \in \mathcal{H}_\ell \right\},$$

and define the fixed union $\mathcal{F}_\mathfrak{C} := \bigcup_{\ell=1}^L \mathcal{F}_\ell$. Every function in this union is measurable and takes values in $[0, B_\mathfrak{C}]$. Apply Equation (38) once to $\mathcal{F}_\mathfrak{C}$, using lemma 7 exactly as in the proof of theorem 8. With probability at least $1 - \delta$, simultaneously for every ℓ and $h \in \mathcal{H}_\ell$,

$$\mathcal{R}_{w, \ell}(h) \leq \widehat{\mathcal{R}}_{N, w, \ell}(h) + \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\mathfrak{C}) + 3B_\mathfrak{C} \sqrt{\frac{\log(4/\delta')}{2\mu}}. \quad (42)$$

Condition on $\mathcal{S}_\mu$ and define

$$Z_\ell(\sigma) := \sup_{f \in \mathcal{F}_\ell} \left| \frac{2}{\mu} \sum_{j=1}^\mu \sigma_j f(\widetilde{W}_j) \right|.$$

Changing one Rademacher sign changes Z_ℓ by at most $4B_{\mathfrak{C}}/\mu$. The bounded-difference exponential inequality (McDiarmid, 1989) therefore gives, for every $t > 0$,

$$\mathbb{E}_{\sigma} \exp(t[Z_\ell - \mathbb{E}_{\sigma} Z_\ell]) \leq \exp\left(\frac{2t^2 B_{\mathfrak{C}}^2}{\mu}\right).$$

The log-sum-exp inequality then gives

$$\mathbb{E}_{\sigma} \max_{\ell \in [L]} Z_\ell \leq \max_{\ell \in [L]} \mathbb{E}_{\sigma} Z_\ell + \frac{\log L}{t} + \frac{2t B_{\mathfrak{C}}^2}{\mu}.$$

Optimizing at $t = \sqrt{\mu \log L / (2B_{\mathfrak{C}}^2)}$ (with the $L = 1$ case immediate) yields the standard finite-union estimate (Bartlett & Mendelson, 2002)

$$\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_{\mathfrak{C}}) \leq \max_{\ell \in [L]} \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\ell) + 2B_{\mathfrak{C}} \sqrt{\frac{2 \log L}{\mu}}.$$

The preceding closed-form calculation applies candidate by candidate, so

$$\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\ell) \leq \frac{8\Lambda_\ell(\Lambda_\ell + \Upsilon) + 2\Upsilon^2}{\sqrt{\mu}} \leq \frac{A_{\mathfrak{C}}}{\sqrt{\mu}}.$$

Substitute these bounds into Equation (42) and apply the candidate-specific inequality $\mathcal{R}_{\infty, \ell}(h) \leq \mathcal{R}_{w, \ell}(h) + \tau_{w, \ell}$. This proves Equation (30). Because the event is simultaneous over the complete fixed union, it remains valid after any measurable sample-dependent selector taking values in that union. $\square$

B.11 Geometrically mixing case

Proof. Since $\mu = N/(2a)$ and $a \geq 1$,

$$\begin{aligned} 4(\mu - 1)\beta_Z(as - w) &\leq \frac{2N}{a}\beta_0 e^{-c(as-w)} \\ &\leq 2N\beta_0 e^{cw} e^{-cas}. \end{aligned}$$

The choice in Equation (31) makes the last expression at most $\delta/2$, so $\delta' \geq \delta/2$. It also gives $a = O(\log(N/\delta))$ with the remaining quantities fixed. Since $\mu = N/(2a)$, both stochastic terms in Equation (29) are $O(\sqrt{a/N}) = O(\sqrt{\log(N/\delta)/N})$.

For a nondivisible available sample size, retain $N_a = 2a \lfloor N/(2a) \rfloor$ observations. When $N_a \geq 2a$, stationarity permits applying the same bound to this retained consecutive segment, and the number discarded is at most $2a - 1$. For fixed parameters, the chosen $a = O(\log N)$, so $a \leq N/2$ and $N_a \geq 2a$ for all sufficiently large N . $\square$

B.12 Local-Pauli shadow requirement

Proof. For independently uniform local $X/Y/Z$ measurements, a nonidentity Pauli string of weight $q \leq k$ has squared shadow norm $3^q \leq 3^k$. The median-of-means classical-shadow guarantee applied to M observables therefore gives

$$S \geq C_{\text{sh}} 3^k \varepsilon^{-2} \log(2M/\delta_{\text{sh}})$$

for a universal constant C_{sh} ; see Huang et al. (2020). For NR reservoir output states, assign failure probability $\delta_{\text{sh}} = \delta_{\text{tot}}/(NR)$ to each state and take a union bound. This gives Equation (33). Each state requires S independently randomized snapshots, for NRS state preparations. Since each snapshot replays the w -step recursion, the total is $NRSw$ channel applications. $\square$

B.13 Finite-shot feature perturbations

Proof. The coordinate-wise event gives $\|\hat{\Phi}_w - \Phi_w\|_2 \leq \sqrt{R|\mathcal{O}|}\varepsilon$. The following argument is deterministic and uses only $\|h\|_{\mathcal{H}_\kappa} \leq \Lambda$, so it holds simultaneously for every $h \in \mathcal{H}_{\kappa,\Lambda}$. The reproducing property and lemma 4 yield $\left| h(\hat{\Phi}_w) - h(\Phi_w) \right| \leq \Lambda \left\| \kappa(\hat{\Phi}_w, \cdot) - \kappa(\Phi_w, \cdot) \right\|_{\mathcal{H}_\kappa} \leq (\Lambda/\xi) \sqrt{\nu R|\mathcal{O}|/(\nu-1)}\varepsilon$, proving Equation (34). Both predictions lie in $[-\Lambda, \Lambda]$; the squared-loss Lipschitz inequality used in the proof of corollary 5 therefore gives Equation (35). $\square$

C Theory-Aligned Dependent Synthetic Families

The controlled experiments use synthetic processes for which stationarity, beta mixing, and bounded labels can be justified from the data-generating mechanism. The three families below deliberately represent different sources of temporal dependence: linear vector dynamics, latent regime persistence, and conditional heteroskedasticity. They are not intended to exhaust the processes covered by theorem 8; rather, they provide a compact set of qualitatively distinct and analytically traceable test beds.

C.1 Finite-block supervised transformations

We first record a closure property used for all three constructions. It separates the mixing argument for the input process from the choice of bounded supervised target.

Lemma 14 (Finite-block transformations preserve beta mixing; cf. Bradley, 2005). *Let $(V_t)_{t \in \mathbb{Z}}$ be a stationary beta-mixing process, let $(E_t)_{t \in \mathbb{Z}}$ be an i.i.d. process independent of V , and let ψ and f be measurable maps. For integers $\ell_-, \ell_+ \geq 0$, define*

$$X_t = \psi(V_t), \quad Y_t = f(V_{t-\ell_-:t+\ell_+}, E_t), \quad Z_t = (X_t, Y_t).$$

Then Z is stationary and, for every $q > \ell_- + \ell_+$,

$$\beta_Z(q) \leq \beta_V(q - \ell_- - \ell_+). \quad (43)$$

If $|f| \leq \Upsilon$, then the bounded-label condition in Section 3.4 also holds.

Proof. Let $U_t = (V_t, E_t)$. Because E is i.i.d. and independent of the complete process V , adjoining the two disjoint innovation sigma-algebras does not increase the absolute-regularity coefficient; this is the standard independent augmentation property of beta mixing (Bradley, 2005). The past sigma-algebra of Z is contained in $\sigma(U_t : t \leq \ell_+)$, while its future sigma-algebra at lag q is contained in $\sigma(U_t : t \geq q - \ell_-)$. Their raw-time separation is $q - \ell_- - \ell_+$. Monotonicity under sigma-algebra inclusion and measurable maps therefore yields

$$\beta_Z(q) \leq \beta_U(q - \ell_- - \ell_+) = \beta_V(q - \ell_- - \ell_+).$$

Stationarity follows because the joint process (V, E) is stationary and the construction is shift equivariant. $\square$

The complete controlled task suite is defined separately in Section D. All shared tasks are bounded finite-block maps of the underlying process, with one future step used only for the genuine forecasting target. The HMM filtering score and GARCH conditional-volatility target are also bounded measurable functions of the corresponding finite-dimensional state constructions. Their compatibility with the assumptions of theorem 8 is summarized in lemma 18.

C.2 Stable vector VARMA

Let (U_t) satisfy the vector VARMA(p, q) recursion

$$U_t = \sum_{i=1}^p A_i U_{t-i} + \varepsilon_t + \sum_{j=1}^q B_j \varepsilon_{t-j}, \quad (44)$$

where the innovations are i.i.d. in $\mathbb{R}^d$. The input exposed to the learner is the bounded coordinatewise transformation

$$X_t = \tanh(U_t/c_x), \quad (45)$$

with a fixed scale $c_x > 0$.

Lemma 15 (Stable vector ARMA processes are geometrically beta mixing; [Mokkadem, 1988](#)). *Assume that the autoregressive polynomial $A(z) = I - \sum_{i=1}^p A_i z^i$ is nonsingular for every $|z| \leq 1$, that the innovations have an absolutely continuous distribution, and that $\mathbb{E} \log^+(1 + \|\varepsilon_0\|_2) < \infty$. Then the standard causal VARMA series is well defined and stationary, and the geometric complete-regularity result of [Mokkadem \(1988\)](#) gives*

$$\beta_U(q) \leq C_V \rho_V^q$$

for constants $C_V < \infty$ and $\rho_V \in (0, 1)$. Consequently, the bounded process in Equation (45), together with the bounded task maps in Section D, satisfies the stochastic assumptions of theorem 8, up to the finite lag shift in Equation (43).

The moment condition is mild and is satisfied by the Gaussian and Student innovations considered in the implementation. The mixing conclusion is the geometric complete-regularity result of [Mokkadem \(1988\)](#); the final statement follows from the data-processing property of beta mixing ([Bradley, 2005](#)) and lemma 14.

Realization used in the controlled suite. We retain the existing $d = 3$, $p = q = 3$ generator with centered Gaussian innovations. For every saved configuration, the spectral radius of the companion matrix is checked to be below one and recorded with the generated artifact. The medium-dependence operating point is rescaled to companion spectral radius 0.7; additional sensitivity points use 0.4 and 0.9. The same fixed matrices, innovation covariance, scaling c_x , and target vectors are reused across all compared methods for a given seed.

High-dimensional extension. For the ambient-dimension experiment in Section 6.5, the same stable construction is generated at $d \in \{3, 10, 30, 100, 300, 500\}$ with medium dependence and identical sample sizes. The quantum register width is not set equal to d . Sparse and distributed task directions are defined in Section D.4; they separate added nuisance coordinates from labels whose signal is spread across all input dimensions.

C.3 Finite-state hidden Markov observations

Let (S_t) be a Markov chain on $[K] = \{1, \dots, K\}$ with transition matrix P . Conditional on $S_t = j$, draw an emission

$$U_t = \mu_j + L_j \varepsilon_t, \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d), \quad X_t = \tanh(U_t), \quad (46)$$

where the emission noises are independent of the latent chain. This is the standard finite-state hidden Markov construction of [Baum & Petrie \(1966\)](#).

Lemma 16 (Positive finite-state hidden Markov models are geometrically beta mixing; [Dobrushin, 1956](#); [Bradley, 2005](#)). *Suppose that S_0 has the stationary distribution of P and*

$$\epsilon_P := \min_{i,j \in [K]} P_{ij} > 0.$$

Then the latent chain and the observed process in Equation (46) are stationary and geometrically beta mixing. In particular,

$$\beta_X(q) \leq \beta_S(q) \leq (1 - K\epsilon_P)^q. \quad (47)$$

After adjoining the bounded shared or filtered-regime targets in Section D, the resulting supervised process satisfies the assumptions of theorem 8 with the lag adjustment from lemma 14.

Proof. The Dobrushin coefficient satisfies

$$\delta(P) := \sup_{i,i'} \|P(i, \cdot) - P(i', \cdot)\|_{\text{TV}} \leq 1 - K\epsilon_P.$$

For a stationary finite-state Markov chain, $\beta_S(q) = \sum_i \pi_i \|P^q(i, \cdot) - \pi\|_{\text{TV}}$. Dobrushin contraction therefore gives $\beta_S(q) \leq \delta(P)^q \leq (1 - K_{\epsilon_P})^q$ (Dobrushin, 1956; Bradley, 2005). Write the emission as $X_t = g(S_t, E_t)$ with i.i.d. E_t independent of S . Independent augmentation and measurable data processing give $\beta_X(q) \leq \beta_{(S,E)}(q) = \beta_S(q)$. The supervised conclusion then follows from lemma 14. $\square$

Realization used in the controlled suite. We use $K = d = 3$, initialize the latent chain from its stationary (discrete-uniform) distribution, and set

$$P_\varrho = \varrho I_3 + \frac{1-\varrho}{2}(\mathbf{1}\mathbf{1}^\top - I_3), \quad \varrho \in \{0.60, 0.80, 0.95\}. \quad (48)$$

The three emission means and covariances are

$$\mu_1 = (-1.5, 0, 1.0)^\top, \quad \mu_2 = (0.5, 1.5, -1.0)^\top, \quad \mu_3 = (1.5, -1.0, 0.5)^\top, \quad L_j = 0.35I_3.$$

The middle persistence level $\varrho = 0.80$ is the main operating point; 0.60 and 0.95 test faster and slower regime switching. Because all entries of P_ϱ are positive, every realization satisfies lemma 16.

C.4 Gaussian GARCH(1,1)

To study volatility clustering, let

$$R_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \omega + \alpha_G R_{t-1}^2 + \beta_G \sigma_{t-1}^2, \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad (49)$$

which is the GARCH(1,1) model introduced by Bollerslev (1986). We expose three bounded return-derived coordinates to the learner,

$$X_t = (\tanh(R_t/2), \tanh(|R_t|/2), \tanh(R_t^2/4))^\top. \quad (50)$$

The observation map exposes signed return, magnitude, and squared-return information while keeping the learner input bounded. The family-specific prediction target is the conditional-volatility state in Equation (62).

Lemma 17 (Gaussian GARCH(1,1) is geometrically beta mixing under a moment contraction; Bougerol & Picard, 1992; Carrasco & Chen, 2002). *Assume $\omega > 0$, $\alpha_G > 0$, $\beta_G \geq 0$, Gaussian innovations, and that for some $s > 0$,*

$$\mathbb{E}[(\alpha_G \varepsilon_0^2 + \beta_G)^s] < 1. \quad (51)$$

Then the stationarity conditions of Bougerol & Picard (1992) and the Gaussian-density regularity conditions used by Carrasco & Chen (2002) are satisfied: the GARCH state has a unique strictly stationary solution and is geometrically beta mixing. Hence the bounded observation process in Equation (50), together with the shared tasks and the conditional-volatility target in Section D, satisfies the stochastic assumptions of theorem 8 after the lag adjustment in lemma 14.

Condition Equation (51) is a convenient sufficient specialization of the cited stationarity and mixing results. It implies $\mathbb{E} \log(\alpha_G \varepsilon_0^2 + \beta_G) < 0$ by Jensen’s inequality. In the Gaussian realization, the transparent condition $\alpha_G + \beta_G < 1$ implies Equation (51) with $s = 1$.

Realization used in the controlled suite. We set $\omega = 1 - \alpha_G - \beta_G$, so the stationary second moment is one, and use

$$(\alpha_G, \beta_G) \in \{(0.10, 0.60), (0.10, 0.80), (0.07, 0.90)\}. \quad (52)$$

These settings have persistence sums 0.70, 0.90, and 0.97, respectively; the middle setting is used in the main cross-family comparison. The other two provide low- and high-persistence sensitivity checks. All three satisfy lemma 17 with $s = 1$.

C.5 Common dataset realization and empirical role

The cross-family robustness suite uses the same windowing and split protocol for all three generators. For each predeclared data-generating seed, we form $N_{\text{tr}} = 5000$ training windows and $N_{\text{te}} = 1000$ test windows with $w = 25$ and stride $s = 100$. The first 5000 windows form the training portion and the final 1000 form a chronological test portion; the final 20% of the training portion is reserved for inner validation. Five data-generating seeds are fixed before experiments are run. HMM trajectories start exactly in stationarity. For the Gaussian VARMA realization, we use its linear state-space representation, solve the associated discrete Lyapunov equation, and draw the augmented initial state from the resulting stationary Gaussian law. For GARCH, the theorem concerns the ideal stationary solution; the numerical generator approximates that law using an independent burn-in of 10^5 raw time steps. This approximation is kept identical across methods and is reported as a simulation approximation, not as proof of exact stationarity.

Table 3: Controlled dependent-family suite. The middle persistence setting is used in the main comparison; the remaining settings probe sensitivity to the strength of temporal dependence.

Family	Dependence mechanism	Persistence settings			Main targets
VARMA	Linear vector recurrence	companion	radius	in	shared suite; high-dimensional variants
		$\{0.4, 0.7, 0.9\}$			
HMM	Latent regime switching	$\varrho \in \{0.60, 0.80, 0.95\}$			exponential, Volterra, filtered regime
GARCH	Conditional variance recursion	$\alpha_G + \beta_G \in \{0.70, 0.90, 0.97\}$			exponential, Volterra, conditional volatility

The detailed functional, architecture, high-dimensional, and shot studies remain concentrated on VARMA to avoid multiplying every experiment by three. HMM and GARCH are used for the cross-family experiment with the same selected QuaRK configuration, the shared exponential and Volterra tasks, one family-specific task, and the principal matched classical controls. The exact allocation is given in Table 4.

D Empirical Temporal Functionals

The controlled experiments separate the dynamics that generate the inputs from the temporal map that generates the response. This appendix defines the task family used throughout the synthetic study and fixes its normalization, delays, and allocation across experiments. The hierarchy follows the usual reservoir-computing distinction between linear memory and nonlinear information processing (Jaeger, 2001; Dambre et al., 2012), while the exponentially weighted and Volterra-type tasks are finite-window versions of classical fading-memory operators (Boyd & Chua, 1985).

D.1 Notation and common normalization

Let

$$\mathbf{X}_{t,w} := (X_{t-w+1}, \dots, X_t) \in ([-1, 1]^d)^w$$

be the input window presented to the learner. Choose directions $u_d, v_d \in \mathbb{R}^d$ satisfying $\|u_d\|_1, \|v_d\|_1 \leq 1$, and fix the memory-decay parameter $\gamma = 0.8$. For the base dimension $d = 3$, the common directions are

$$u_3 = \frac{1}{3}(1, -1, 1)^\top, \quad v_3 = \frac{1}{3}(1, 1, -1)^\top.$$

All task directions, delays, decay parameters, and any task-generation seeds are fixed before the data are simulated; all methods evaluated on a given dataset receive the same realized task. Define

$$A_w := \sum_{j=0}^{w-1} \gamma^j = \frac{1 - \gamma^w}{1 - \gamma}. \quad (53)$$

The normalizations below place every controlled target in $[-1, 1]$, so the bounded-label constant can be taken as $\Upsilon = 1$ for this suite.

D.2 Shared temporal tasks

One-step future projection. The genuine forecasting task uses one observation beyond the input window,

$$F_{\text{fore}}(\mathbf{X}_{t,w}, X_{t+1}; u_d) := u_d^\top X_{t+1}. \quad (54)$$

The learner receives observations only through time t ; X_{t+1} is used solely to construct the label. This distinguishes the task from an instantaneous decoding check based on X_t .

Exponentially weighted linear memory. Let

$$L_u(t) := \sum_{j=0}^{w-1} \gamma^j u_d^\top X_{t-j}. \quad (55)$$

The normalized fading-memory target is

$$F_{\text{exp}}(\mathbf{X}_{t,w}; u_d) := \frac{L_u(t)}{A_w}. \quad (56)$$

It requires information from the full window while assigning greater weight to recent observations.

Second-order Volterra-type memory. With $L_v(t)$ defined as in Equation (55) using v_d , set

$$F_{\text{vol}}(\mathbf{X}_{t,w}; u_d, v_d) := \frac{L_u(t) + \frac{1}{2} L_v(t)^2}{A_w + \frac{1}{2} A_w^2}. \quad (57)$$

This task combines distributed linear memory with a smooth nonlinear interaction across lags.

Delayed recall. For a delay $\tau \in \{1, 5, 10, 20\} \subset \{0, \dots, w-1\}$, define

$$F_{\text{delay}, \tau}(\mathbf{X}_{t,w}; u_d) := u_d^\top X_{t-\tau}. \quad (58)$$

The resulting delay curve is a direct diagnostic of how much information about a particular past observation remains in the final reservoir feature vector. It is interpreted as an empirical memory profile, not as an estimator of the worst-case contraction bound.

Sparse cross-lag interaction. For the predeclared delays $(\tau_1, \tau_2) = (0, 15)$, define

$$F_{\text{cross}}(\mathbf{X}_{t,w}; u_d, v_d) := (u_d^\top X_{t-\tau_1})(v_d^\top X_{t-\tau_2}). \quad (59)$$

Unlike the dense Volterra target, this functional isolates one nonlinear interaction between a recent and a distant observation.

D.3 Family-specific tasks

The common tasks make performance comparable across generators. Two additional targets exploit the defining latent structure of the HMM and GARCH families.

HMM filtered regime score. For the three-state HMM in Section C.3, assign state scores $a = (-1, 0, 1)^\top$. Using the known generator parameters, compute the finite-window filtering probabilities

$$\pi_{t,w}(j) := \mathbb{P}(S_t = j \mid X_{t-w+1:t}), \quad j \in \{1, 2, 3\}, \quad (60)$$

by the standard forward recursion, initialized from the stationary state probabilities at the left edge of the window (Baum & Petrie, 1966). The target is the posterior regime score

$$F_{\text{HMM}}(\mathbf{X}_{t,w}) := \sum_{j=1}^3 a_j \pi_{t,w}(j). \quad (61)$$

The learner receives only the emission window; neither the latent state nor the filtering probabilities are included in its input.

GARCH conditional-volatility state. For the GARCH process in Section C.4, let $\bar{\sigma}^2 := \omega/(1 - \alpha_G - \beta_G)$ be the stationary second moment under the finite-variance parameterization (Bollerslev, 1986). The bounded target

$$F_{\text{GARCH},t} := \tanh \left[c_\sigma \log \left(\frac{\sigma_{t+1}^2}{\bar{\sigma}^2} \right) \right], \quad c_\sigma = \frac{1}{2}, \quad (62)$$

asks the learner to recover the next conditional-volatility state from the bounded return-derived input window. The latent variance is used only to construct the synthetic label.

D.4 Sparse and distributed high-dimensional targets

The ambient-dimension study distinguishes two ways in which the task may scale with d . For $d \geq 3$, define sparse directions by embedding the three-dimensional task,

$$u_d^{\text{sp}} = (u_3^\top, 0, \dots, 0)^\top, \quad v_d^{\text{sp}} = (v_3^\top, 0, \dots, 0)^\top. \quad (63)$$

Here the added coordinates are nuisance dimensions. To distribute signal over all coordinates, let $s^{(u)}, s^{(v)} \in \{-1, +1\}^{\mathbb{N}}$ be two fixed Rademacher sign sequences generated once from predeclared task seeds 1729 and 2718, and set

$$u_d^{\text{de}} := \frac{1}{d} (s_1^{(u)}, \dots, s_d^{(u)})^\top, \quad v_d^{\text{de}} := \frac{1}{d} (s_1^{(v)}, \dots, s_d^{(v)})^\top. \quad (64)$$

Both constructions retain unit ℓ_1 norm and hence a dimension-independent label range. The high-dimensional experiment evaluates F_{exp} under both sparse and distributed directions and F_{vol} under the distributed directions.

D.5 Assumption compatibility

Lemma 18 (The controlled functional suite satisfies the bounded-label and mixing requirements). *For each process realization in Section C, all targets in Equations (54), (56) to (59), (61) and (62) are measurable and bounded by one in absolute value. When paired with the corresponding input process, the resulting supervised process is stationary and beta mixing under the conditions of lemmas 15 to 17.*

Proof. The ℓ_1 -normalization of the directions and $X_t \in [-1, 1]^d$ give the bounds for forecasting, delayed recall, and the cross-lag product. Equation (53) and the denominators in Equations (56) and (57) give the same bound for the two fading-memory targets. The HMM score is a convex combination of values in $[-1, 1]$, and the GARCH target is bounded by the outer hyperbolic tangent. Each common task is a finite-block function of the underlying process. The forecast target uses one future observation and therefore invokes lemma 14 with $\ell_+ = 1$; the remaining shared tasks have $\ell_+ = 0$. The HMM score is a measurable function of a finite emission block. Although it is indexed by $t + 1$, $\sigma_{t+1}^2 = \omega + \alpha_G R_t^2 + \beta_G \sigma_t^2$ is a measurable function of the current stationary GARCH state. The conclusion follows from lemma 14 and the family-specific mixing lemmas. No unbounded label noise is added in the theory-aligned controlled suite. $\square$

D.6 Task allocation across the empirical study

The full Cartesian product of generators and targets is intentionally avoided. Table 4 assigns each functional to a specific scientific question.

Table 4: Compact allocation of controlled functionals. “Shared” means that the same mathematical target is applied to each listed generator; the family-specific targets use latent variables only to construct labels.

Experiment	Generators	Functionals
Functional hierarchy and learning curves	VARMA, medium persistence	future projection, exponential memory, Volterra, delay curve, cross-lag
Memory and persistence study	VARMA, HMM, GARCH at three persistence levels	exponential memory with several decays and delayed recall under a contraction sweep
Cross-family robustness	VARMA, HMM, GARCH, medium persistence	shared exponential and Volterra tasks; VARMA future projection; HMM filtered score; GARCH volatility state
Ambient-dimension study	VARMA with $d \in \{3, 10, 30, 100, 300, 500\}$	sparse and distributed exponential memory; distributed Volterra
Resource and memory ablations	VARMA, medium persistence	Volterra and delayed recall at $\tau \in \{10, 20\}$
Finite-shot study	VARMA, medium persistence	exponential memory, Volterra, and delayed recall at $\tau = 10$
Chaotic stress tests	Lorenz-63 and Mackey-Glass	multi-horizon forecasting and partial-observation prediction; see Section E
Real-world sanity checks	three TSER datasets	archive-provided scalar regression targets; see Section F

E Out-of-Scope Chaotic Stress Tests

The main analysis concerns stationary beta-mixing input–output processes. The experiments in this appendix deliberately leave that scope and ask whether the architecture remains useful on deterministic systems with sensitive dependence on initial conditions. The systems are therefore empirical stress tests, not examples covered by theorem 8. We use the Lorenz-63 flow (Lorenz, 1963) and the Mackey-Glass delay equation (Mackey & Glass, 1977), both of which are standard nonlinear temporal benchmarks.

E.1 Lorenz-63 system

The Lorenz-63 state $U(t) = (u_1(t), u_2(t), u_3(t))$ satisfies

$$\begin{aligned} \dot{u}_1 &= \sigma(u_2 - u_1), \\ \dot{u}_2 &= u_1(\rho - u_3) - u_2, \\ \dot{u}_3 &= u_1 u_2 - \beta u_3, \end{aligned} \quad (\sigma, \rho, \beta) = \left(10, 28, \frac{8}{3}\right). \quad (65)$$

Trajectories are generated with a fourth-order Runge-Kutta integrator using step size $h_{\text{int}} = 0.01$, followed by a burn-in of 5000 integration steps. To vary the effective prediction difficulty, the retained trajectory is sampled at

$$\Delta t \in \{0.02, 0.05, 0.10\}.$$

The observed input is the bounded state

$$X_j = \tanh(D^{-1}U(j\Delta t)),$$

where the diagonal scale D is fixed from a separate calibration trajectory and is not estimated from the test segment.

For a unit vector $u \in \mathbb{R}^3$, the scalar q -step target is

$$Y_{j,q}^{\text{L63}} := u^\top X_{j+q}, \quad q \in \{1, 5, 10, 20\}. \quad (66)$$

A second partial-observation task gives the learner only $(X_j^{(1)}, X_j^{(2)})$ and asks it to predict the bounded third coordinate at horizon q . This separates ordinary full-state forecasting from recovery of an unobserved component.

E.2 Mackey–Glass delay system

The scalar Mackey–Glass process is defined by the delay differential equation

$$\dot{x}(t) = \frac{\beta_{\text{MG}} x(t - \tau)}{1 + x(t - \tau)^{n_{\text{MG}}}} - \gamma_{\text{MG}} x(t), \quad (67)$$

with

$$\beta_{\text{MG}} = 0.2, \quad \gamma_{\text{MG}} = 0.1, \quad n_{\text{MG}} = 10, \quad \tau \in \{17, 30\}.$$

The delay equation is integrated with step size 0.1, using a constant positive history on $[-\tau, 0]$, discarding the first 5000 integration steps and sampling every ten integration steps. The bounded observation is $X_j = \tanh(c_x(x_j - m_x))$, where m_x and c_x are fixed from a separate calibration trajectory. The forecast targets are

$$Y_{j,q}^{\text{MG}} := X_{j+q}, \quad q \in \{1, 5, 10, 20\}. \quad (68)$$

The two delay values provide different temporal-complexity regimes without changing the readout task.

E.3 Stress-test protocol

For both systems, five predeclared initial-condition seeds are used. Each trajectory is split chronologically into training, validation, and test segments with a guard interval of at least the largest prediction horizon between adjacent segments. Windows never cross a split boundary. Observation noise is added after trajectory generation at levels

$$\sigma_{\text{obs}} \in \{0, 0.01, 0.05\}$$

and is generated independently across methods from shared seeds.

The principal methods are QUARK, ESN plus Matérn KRR, and matched-dimensional random nonlinear features plus Matérn KRR. Candidate families and readout grids are predeclared exactly as in Section 6.2. We report NRMSE by horizon, the relative skill against a persistence predictor, and the horizon at which that skill falls below a predeclared threshold. No mixing coefficient, effective-block count, or theorem-based confidence statement is attached to these results.

Table 5: Chaotic stress-test matrix.

System	Controlled variations	Tasks
Lorenz–63	$\Delta t \in \{0.02, 0.05, 0.10\}$, three noise levels	full-state scalar forecasting and partial-observation third-coordinate prediction, $q \in \{1, 5, 10, 20\}$
Mackey–Glass	$\tau \in \{17, 30\}$, three noise levels	scalar forecasting at $q \in \{1, 5, 10, 20\}$

F Limited Real-World Sanity Checks

The main empirical contribution is the controlled synthetic characterization. This appendix adds a deliberately small external check using three datasets from the Monash–UEA–UCR Time Series Extrinsic Regression archive (Tan et al., 2021). The datasets were chosen before inspecting the final comparison table to span markedly different temporal lengths and channel counts. They are not used to claim broad real-world validation, and the stationarity and beta-mixing assumptions of the theory are not asserted for them.

F.1 Datasets and roles

Table 6: Real-world sanity-check datasets. The dimensions and audited splits match the repository manifests used by the empirical pipeline.

Dataset	Train	Test	w	d	Role in the sanity check
Gas Sensor Array Ethanol	324	140	7500	1	Very long univariate sensor trace; tests compression of long temporal context.
Live Fuel Moisture Content	3000	1000	365	7	Year-long multivariate remote-sensing series; tests moderate channel count and seasonal context.
Hydraulic Systems	1159	290	60	728	Short but very high-dimensional multivariate record; tests the fixed-width projection pipeline under large raw dimension.

The archive defines a time-series extrinsic-regression target for each dataset: a scalar response associated with the entire observed series rather than necessarily a future value (Tan et al., 2021). No synthetic labels or latent variables are introduced. The original archive split is retained, and all preprocessing parameters are fitted on the training portion only.

F.2 Comparison protocol

The sanity check compares:

1. QUARK with the selected resource-aware configuration;
2. an ESN representation of the same reported feature dimension followed by the identical Matérn KRR readout;
3. matched-dimensional random nonlinear features followed by the same readout; and
4. raw-window Matérn KRR when the kernel computation satisfies the predeclared memory and runtime guardrails.

For stochastic representations, five method seeds are paired across methods. The Matérn candidate grid, validation fraction, selection metric, and target normalization are shared. The test set is evaluated only after the readout and all representation hyperparameters have been selected from the training portion.

The primary metric is training-standard-deviation-normalized RMSE. We also report unnormalized MSE, feature dimension, projection time, feature-extraction time, readout time, and peak memory. Dataset-level values and paired seed differences are shown in this appendix. The main text may summarize the three outcomes, but it does not report an average rank or general superiority claim from such a small sample.

F.3 Interpretation boundary

These datasets answer only a limited transfer question: do the qualitative comparisons and failure modes observed in the controlled study remain visible on three naturally collected problems with very different shapes? A favorable result does not establish practical deployability or robustness across the TSER archive. An unfavorable result is equally informative because it identifies where the controlled synthetic conclusions fail to transfer.